

Project 2 Specification Stage — transcript

- **Project:** ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator
- **Repository:** <https://github.com/rg-sureshkumar-goat/AI-Footprint-Calculator>
- **Branch:** project-2 (HEAD at start of session: 067026c; at save: 067026c)
- **Date:** 2026-09-26
- **Learner:** RG Sureshkumar
- **Agent:** Claude Code (Opus 5.5)
- **Stage command:** start specification (instructions in spec.md, Commands section)
- **Outcome:** Specification recorded in spec.md (goal with two side-by-side totals, five features with calculations, limitations, interface expectations, and acceptance checks, out of scope, and three revision notes) and explicitly approved by the learner; sources S59–S80 added to research.md and marked as not yet assessed by the learner.

Note on redaction. Sanitised following transcripts/README.md: the author's email address, macOS home path and username, and temporary/session paths are replaced with placeholders. No messages were removed or reworded.

Note on tool calls. Each tool call is shown as a one-line summary in square brackets, followed by its output truncated to 400 characters. The agent's internal reasoning is not included. The complete local session log is kept locally and not committed.

Note on the final turn. The last learner turn (save transcript) is followed by the tool calls that generated this file; the agent's closing confirmation was written after the file was generated and so is not included.

1. Learner (RG Sureshkumar)

start specification

Agent

[Tool: Bash — Read spec, brief, and list files]

```
Exit code 1
# Technical Specification

> EDITING DIRECTIVE: USER AND AGENT EDIT THIS FILE COLLABORATIVELY. THE USER MUST REVIEW
AND APPROVE ITS CONTENT.

Purpose of this file: Define what the completed project must do so it can be planned,
built, and verified.

## Instructions for the user
```

Translate the approved research into a specification without distorting its evidence, limitations, or uncert ... [truncated]

[Tool: Bash — Read brief and research documents]

<persisted-output>

Output too large (99.5KB). Full output saved to: <session-log>

Preview (first 2KB):

Project 2 Brief

> EDITING DIRECTIVE: DO NOT EDIT THIS FILE. IT IS THE AUTHORITATIVE PROJECT FRAME.

Roles in the project documents: The user is the person directing and working with the AI agent. The agent supports the user's work but does not make or approve decisions on the user's behalf. E ... [truncated]

[Tool: Read —]

1 # Project 2 Brief

2

3 > EDITING DIRECTIVE: DO NOT EDIT THIS FILE. IT IS THE AUTHORITATIVE PROJECT FRAME.

4

5 Roles in the project documents: The user is the person directing and working with the AI agent. The agent supports the user's work but does not make or approve decisions on the user's behalf. Employees are the intended audience for the completed calculator, and the named reference profi ... [truncated]

[Tool: Bash — Show research headings and selected features]

7:## Instructions for the user

11:## Instructions for the agent

17:## Reference employee profiles

25:## Audience needs

31:### 1. Image generation (Alex)

52:### 2. Video generation (Alex)

79:### 3. Coding agents and long, irregular sessions (Jordan)

112:### 4. Occasional text AI (Robin)

141:### 5. Video calls (Jordan, Robin)

163:### 6. Video and music streaming (Alex, Robin)

185:### 7. Social media ... [truncated]

[Tool: Bash — List code files and feature init functions]

assets/css:

base.css

features.css

assets/js:

app.js

```
budget.js
calculator.js
core.js
data.js
nav.js
store.js
team-data.js
team.js
tracker.js

transcripts:
README.md
debrief-2026-09-12_000739.md
debrief-2026-09-12_000739.pdf
research-2026-09-26_180854.md
research-2026-09-26_180854.pdf
session-2026-09-12.md
    31 assets/js/app.js
    296 assets/js/budget.js
    422 assets/js/calculator.js
    ... [truncated]
```

I've read the brief, the approved research, and the spec template. Next I'm checking how the calculator currently produces its total, so the first question I ask you is grounded in the code.

[Tool: Bash — Skim data model and how totals are computed]

```
/**
 * data.js – reference figures for the calculator.
 *
 * Per-model energy / carbon / water estimates (mean plus EcoLogits' own 95%
 * interval) come from the EcoLogits library v0.10, the engine behind the
 * EcoLogits calculator at huggingface.co/spaces/genai-impact/ecologits-calculator.
 * Every other figure is sourced in the methodology section of index.html.
 *
 * Carried over unchanged fro ... [truncated]
```

Specification stage: orientation

The prerequisite is met. You approved research and the five features on 2026-09-26. The spec now has to turn them into testable requirements:

1. Image generation logging (professional AI)
2. Agent session and project estimator (professional AI)
3. Digital day builder (digital life)
4. Device manufacturing spread over use (digital life)
5. Like-for-like comparisons (strongest remaining need)

How the calculator works now: People add rows of text prompts by model and size. The calculator adds them up with `AIPF.aiDaily()` into one daily AI total (energy, carbon or water) and compares that total with

everyday items and a yearly baseline. The session tracker, team dashboard and monthly budget from Project 1 all use that AI-only total.

Proposed order: First the Goal and the definition of "the final calculation", since the brief says every feature must feed it. Then the five features one at a time, each with its need, behaviour, inputs and outputs, calculation, evidence and uncertainty, interface, and acceptance checks. After that, Out of scope. I'll draft each section for you to review, and nothing counts as approved until you say so.

First question: What should "the final calculation" be?

- **(a) Two totals side by side.** The AI daily total stays as it is. Images and agent sessions are added to it. A separate "digital day" total covers calls, streaming, social media, gaming and device manufacturing. AI is shown next to it, not inside it.
- **(b) One combined daily total.** All of the above becomes one "digital footprint" figure, with AI shown as one labelled part of it.

This choice has knock-on effects. It decides whether the tracker, team dashboard and budget stay AI-only or change. It also affects the neutrality framing: in (b), a share like "AI is X% of your digital day" can make AI look small or large depending on the inputs. Which do you want, or is there another option you have in mind?

2. Learner (RG Sureshkumar)

I want two totals side by side

Agent

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've added a draft Goal section to [spec.md](#), under **Goal**, including your decision on the final calculation:

- **AI total:** the current daily AI total, now also counting generated images (feature 1) and agent sessions (feature 2).
- **Digital-day total:** a separate total for calls, streaming, social media, gaming (feature 3) and device manufacturing (feature 4).
- The two are shown side by side.
- Feature 5 corrects how both totals are compared with everyday items.

Two lines in the draft are my reading of option (a), not things you said directly:

1. **The session tracker, team dashboard and monthly budget stay AI-only.** They would count images and agent sessions, but not the digital-day activities.
2. **AI is never shown as a percentage of the digital-day total.** The draft rules out wording like "AI is 5% of your digital day."

Question: Are both of those right, or do you want to change either one? Once they're settled, I'll move on to feature 1, image generation logging.

3. Learner (RG Sureshkumar)

both are right, let move to feature 1

Agent

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded that you confirmed both points.

Feature 1: image generation logging

The approved feature lets employees log generated images by model type and settings, and those images count toward the AI total. Research identified one central constraint.

What the evidence covers: - **Open models only.** All the per-image figures come from open models on research hardware: about 0.09–4.1 Wh on a consumer GPU (S3), a 1.35 Wh median and 11.5 Wh maximum from 2022–23 models (S1), and about 0.32–0.63 Wh for Stable Diffusion 3 Medium (S2). - **No commercial figures.** None of the tools creative staff are likely to use publish anything. Research found no figures for Midjourney, OpenAI, Google or Adobe Firefly. - **Settings matter.** Doubling resolution raised energy 1.3–4.7x (S3). Doubling diffusion steps roughly doubled it (S2). - **Water and carbon would be derived.** There is no measured water figure per image.

Question: How should an employee choose the "model type" when their actual tool can't be measured?

- **(a) One general image range.** Every image uses the full evidence range with a central value. There's no model choice, only settings. This is the simplest and most honest option, but the range is very wide.
- **(b) A few labelled tiers,** such as "lightweight / typical / heavy". Each tier is tied to a slice of the study data. Employees get some control, but they would have to guess which tier their commercial tool falls into, and the tier boundaries would be our construction rather than a source's.
- **(c) The open models named in the studies,** such as Stable Diffusion 3 Medium or SDXL. This is precise for those models, but most employees don't use them.

Whichever you choose, the tool will state that no commercial image tool publishes figures. Which approach do you want?

4. Learner (RG Sureshkumar)

go with option a

Agent

Next I'm checking S3 directly, to see what baseline settings its per-image figures assume, before I draft any numbers.

[Tool: ToolSearch — select:WebFetch]

(no output)

[Tool: WebFetch — <https://arxiv.org/html/2506.17016>]

```
# Energy Consumption Analysis: Key Figures
```

```
## Default Settings
```

```
The paper does not explicitly state a single default resolution or inference step count used for the "per-image energy figures." Instead, they tested multiple configurations:
```

- ```
- **Resolutions**: 512×512, 768×1024, and 1024×1024
- **Inference steps**: Each model used "their default settings," which vary by design (ranging "from as few a ... [truncated]
```

I checked S3 directly (arXiv HTML) and confirmed the 46× spread across 17 models, from 0.086 to 4.08 Wh per image. The paper also sets up a problem for the "settings" part of feature 1:

- **Each model ran at its own defaults.** Step counts ranged from 2 to 50 or more, and three resolutions were tested (512×512, 768×1024 and 1024×1024). S3 deliberately left step count out of its comparison.
- **The resolution result only covers one jump.** The 1.3–4.7× increase comes from going from 512×512 to 1024×1024. Nothing above 1024 was measured.
- **S1's figures are also at each model's defaults.**

With option (a), the general range (about 0.09–11.5 Wh) already mixes models, resolutions and step counts. Applying a resolution or step multiplier on top of it would partly count the same variation twice. Most commercial tools also don't show employees a step count.

**Question:** How should settings work in feature 1?

- **(i) No setting multipliers.** The employee enters how many images they generated, including discarded variations, and each image uses the general range. The interface explains in plain language that resolution and steps are part of why the range is so wide. This is the most defensible option, but it moves away from the approved wording "by model type and settings".
- **(ii) One resolution choice.** The options would be "small (about 512 px)" and "standard (about 1024 px)". The general range would be treated as the standard case, and "small" would scale it down by the measured 1.3–4.7×. This keeps a settings input, but it rests on one study and on the assumption that the general range represents about 1024 px, which is only partly true.

If you choose (i), I'll add a Revisions note saying the approved "settings" input was dropped and why. Which do you want?

## 5. Learner (RG Sureshkumar)

---

go with option 1

## Agent

---

I'm checking how the calculator currently turns energy into carbon and water, so the image figures would convert the same way.

[Tool: Bash — Inspect metric conversion logic]

```
/**
 * core.js – shared state, formatting, and the carbon / water math.
 *
 * Base units throughout: carbon in grams CO2e, water in liters. Everything the
 * UI renders is derived from these two, so a value can be moved between the
 * calculator, the live tracker, the team view, and the budget without
 * conversion bugs.
 *
 * Ported from Andy Masley's original calculator (CC0).
 */
(function (AIP ... [truncated]
```

[Tool: Bash — Locate spec section headings]

```
7:## Instructions for the user
11:## Instructions for the agent
17:## Goal
37:## Features
42:- its behavior, inputs, and outputs
43:- its calculations, supporting evidence, and uncertainty
44:- its interface expectations and acceptance checks
46:## User approval
50:## Out of scope
54:## Revisions
58:## Commands
```

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

Feature 1 is now drafted in [spec.md](#), under **Features** → **Feature 1**, with the open items marked. I also added a Revisions note explaining that the approved "settings" input was dropped and why.

I checked the code to see how images would fit. Text carbon is calculated as energy × the selected region's grid intensity, and images can use the same method. Text water, however, comes straight from EcoLogits per

model, so image water needs a water intensity we choose ourselves. That's one of the open items, and I'll come back to it later.

**Question:** What central value should one image use? The low end (0.09 Wh) and high end (11.5 Wh) are settled. The AI total, tracker and budget all need a single central number as well as the range. The sourced candidates are:

- **(a) 1.35 Wh per image.** This is the median of the image models in S1. It spans many models, but they are 2022–23 models on an A100, so it may be dated.
- **(b) 0.63 Wh per image.** This is Stable Diffusion 3 Medium at 1024 px and 25 steps, including overhead (S2). It's more recent and on data-centre hardware, but it is one model, and its overhead doubling is described in S2's own methodology as untested for image models.

A midpoint I calculated myself (about 1 Wh) wouldn't come from any source, so I'm not suggesting it unless you want it labelled as our own construction. Which central value do you want?

## 6. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: Edit — spec.md]

[Tool: Bash — Check tracker and page sections for image input placement]

The file spec.md has been updated successfully.

```
207: box.innerHTML = '';
221: box.innerHTML = '';
249: box.innerHTML = '';
297: ms.innerHTML = '';
305: ss.innerHTML = '';
76: <h2 class="aipf-panel-title" id="aipf-session-h"><span class="aipf-pulse" aria-
hidden="true"> Today's session</h2>
129: <h2 class="aipf-h2" id="aipf-usage-h">Your use, against the rest of your life</h2>
181: <h2 class="aipf-panel-ti ... [truncated]
```

[Tool: Bash — Read tracker header and session panel markup]

```
/**
 * tracker.js — FEATURE 1: Live session tracker.
 *
 * The original calculator asks you to describe an average day. That is the
 * wrong question for someone who uses AI all day at work: nobody knows their
 * own average, and the estimate never changes as the day goes on. This logs
 * prompts as they actually happen, keeps a running total that updates live,
 * and projects where the day is hea ... [truncated]
```

I've recorded the 1.35 Wh central value.

The next open item is where the image input goes, and the code matters here. The session tracker has a switch, "Use my log to drive this page", which is on by default. When it's on, today's tracker log replaces the calculator's typed-in estimates for the whole page, including the totals, charts, budget and team view.

That means an image input placed only in the calculator would be ignored whenever the switch is on.

**Question:** Where should employees enter images?

- **(a) Both places.** Images get a count field in the calculator and a "Log image" action in the tracker, so they count whichever mode is active. This is the most complete option, and it keeps images consistent with how text prompts already work.
- **(b) Tracker only.** Images are logged as they're made, like prompts, and don't appear in the typed-in estimate. This is simpler, but images disappear from the page if someone turns the log switch off.
- **(c) Calculator only.** This is the simplest option, but images wouldn't count while the log switch is on, which is the default.

Which do you want?

## 7. Learner (RG Sureshkumar)

---

go with option b

### Agent

---

[Tool: Edit — spec.md]

[Tool: WebFetch — [https://ecologits.ai/latest/methodology/llm\\_inference/](https://ecologits.ai/latest/methodology/llm_inference/)]

The file spec.md has been updated successfully.

```
Water Consumption Calculation in EcoLogits
```

```
Formula
```

EcoLogits computes Water Consumption Footprint (WCF) for the usage phase using this formula:

```
> "WCF^u_request = E_server × [WUE_on-site + PUE × WUE_off-site]"
```

This multiplies server energy by the sum of on-site and off-site water use factors.

```
Water Factors and Sources
```

```
On-site WUE (L/kWh): Assessed individually per AI provider. Sp ... [truncated]
```

[Tool: Bash — Read water source assessments S45–S47]

### S45. Li, Yang, Islam & Ren (2025)

- Citation: Li, P., Yang, J., Islam, M. A., & Ren, S. (2025). Making AI Less "Thirsty": Uncovering and Addressing the Secret Water Footprint of AI Models. \*Communications of the ACM\*, 68(7), 54–61. <<https://doi.org/10.1145/3724499>>; preprint: <<https://arxiv.org/abs/2304.03271>>

- Claim or figure: AI water use spans scope 1 (on-site cooling), scope 2 (electrici ... [truncated]

Recorded: images are logged in the tracker only. The draft states the consequence plainly: with the log switch off, images aren't counted.

Open item: the water intensity for image water

I checked the EcoLogits methodology page directly. It calculates text water as server energy × (on-site cooling water + PUE × off-site electricity water). So the calculator's current text water counts cooling plus power-plant water. The page doesn't publish its actual L/kWh values. Image water should cover the same scope, so that feature 5's like-for-like comparisons hold.

The candidates are:

| Option | Factor                      | Basis                                                                                                                                       | 1.35 Wh image (range 0.09–11.5 Wh) |
|--------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| (a)    | about 4.9 L/kWh             | S46 (Berkeley Lab 2024, rated high confidence): 0.36 L/kWh average on-site plus 4.52 L/kWh through electricity, for US data centres in 2023 | about 6.6 mL (0.4–56 mL)           |
| (b)    | about 3.6–4.1 L/kWh         | The ratio implied by the calculator's own EcoLogits text figures (for example, GPT-5.5 at 9.6 mL for 2.66 Wh)                               | about 4.9–5.5 mL                   |
| (c)    | about 4–12 L/kWh as a range | S45: 1–9 L/kWh on-site plus 3.14 L/kWh for US electricity                                                                                   | about 5–16 mL                      |

Each option has a catch: - **(a)** is a published, sourced factor, but it's US-only and modelled. - **(b)** is the most consistent with the text figures, but we'd be back-calculating it from our own data, so it isn't a published factor. - **(c)** shows the unresolved disagreement about on-site water (research §10), but it widens a range that is already very wide.

Question: Which water intensity should image water use?

8. Learner (RG Sureshkumar)

go with option a

Agent

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

The file spec.md has been updated successfully.

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

Feature 1 is fully drafted in [spec.md](#), under **Features** → **Feature 1**, with no open items left. Here's what it now says:

**Your decisions:** - **Input:** a count of images, including discarded variations. There's no model or settings choice, and images are logged in the tracker only. - **Energy per image:** 1.35 Wh (S1 median), with a range of 0.086–11.49 Wh (S3 low, S1 high). - **Carbon:** energy × the selected region's grid intensity. No embodied carbon is added, and the interface says so. - **Water:** energy × 4.88 L/kWh from S46, labelled as derived.

**Acceptance checks:** With the US region selected, one image adds about **0.51 g CO<sub>2</sub>e (0.033–4.37 g)** and **6.6 mL of water (0.42–56 mL)**. I calculated those from the decided figures.

A few items are my proposals rather than your decisions, so please check them in particular: - **Logging several images at once.** I wrote that the tracker lets employees log more than one image in a single action. - **Same handling as prompts.** Images appear in the recent list and history, and "Undo last" removes them. - **Methodology documentation.** The methodology section of `index.html` documents the image figures and their sources.

**Question:** Does the Feature 1 draft look right, or is there anything to change before we move to feature 2, the agent session and project estimator?

## 9. Learner (RG Sureshkumar)

---

looks right, lets move onto feature 2

## Agent

---

### Feature 2: agent session and project estimator

The approved feature has two goals. It should replace the current "coding / agent session", which counts 100,000 **output** tokens only, with an estimate built from a session's token composition. It should also let employees add sessions up across a project.

**What the evidence gives us:** - **Output tokens:** EcoLogits' per-model energy, already in `data.js` (S9). - **Fresh input tokens:** about 1–4 Wh per 10,000 tokens (S8, S10, S11). - **Cached input tokens:** no evidence-based figure. S8's rate of about 0.39 Wh per 10,000 is a price-based assumption, and its author says he has "no idea" whether it's reasonable. - **Session structure:** input dominates. The same task can vary up to 30x in

tokens (S7). S8's median session was about 592,000 tokens, mostly cache reads, but it's one user's "napkin math" (rated low confidence and illustrative only).

The first decision is the inputs, because everything else depends on them. Many employees won't know their token counts, and Jordan explicitly finds "prompts per day" too simple.

**Question:** How should an employee describe a session?

- **(a) Enter token counts directly.** They'd enter fresh input, cached input and output per session, taken from whatever usage figures their tool reports. This is the most faithful option, but it's unusable for anyone who can't find those numbers.
- **(b) Choose from presets,** such as short, typical or long sessions. This is easy, but the only session-level data is S8's single user, so the presets would rest mostly on low-confidence evidence.
- **(c) Direct token entry, pre-filled with S8's median session as an example.** The example would be clearly labelled as one developer's illustrative data, not a typical value, and employees could overwrite it.

Which approach do you want?

## 10. Learner (RG Sureshkumar)

---

go with option c

### Agent

---

Next I'm checking S8's post for the median session's exact token breakdown, since those numbers will pre-fill the fields.

[Tool: WebFetch — <https://simonpcouch.com/blog/2026-01-20-cc-impact/>]

```
Claude Code Session Energy Breakdown
```

```
Based on the article, here's what the author reports for the median session:
```

```
Token Breakdown:
```

```
The article states: "The median session had 24 requests and consumed 592,439 total tokens—5 user messages and 19 tool call responses." However, the author does not provide the exact breakdown of fresh input tokens, cache write tokens, cache read tokens, and out ... [truncated]
```

[Tool: Bash — Search S8 page text for token breakdown]

```
160: Most of the discourse about the environmental impact of LLM use focuses on a 'median query.' What about a Claude Code session?
```

```
192:That said, 1 or 10 or 100 median prompts a day is many orders of magnitude off from my own personal use of LLMs; I likely am, in Hannah Ritchie's words, an "extreme power user." I work in software and spend much of my workday driving 2 or 3 coding agent ... [truncated]
```

I read S8 directly. There's a problem with option (c).

**S8 doesn't publish the median session's token breakdown.** It gives the total (592,439 tokens), the 24 requests, the 5 user messages and 19 tool-call responses, and the resulting ~41 Wh. It never says how the tokens split between fresh input, cached input and output. The only per-type numbers in the post are a few individual API calls from a data preview, not the median.

So I can't pre-fill the three fields from S8 without making up a split, and I won't do that.

The same post does confirm one useful fact. Claude Code's session logs record four token types per API call: input, output, cache creation (writes) and cache read. So a real tool does expose the kind of data option (c) relies on.

**Question:** How should we handle the missing breakdown?

- **(i) No pre-filled values.** The fields start empty. Next to them, a labelled reference note describes S8's median session: about 592,000 tokens, mostly cache reads, 24 model calls, about 41 Wh. Employees get a sense of scale without the tool presenting invented numbers as data.
- **(ii) Pre-fill only the total.** The field shows about 592,000 tokens, and the split is left for the employee to fill in. This is awkward, because the total alone can't be calculated without a split.
- **(iii) Look for a different published session with a full breakdown** before deciding. I'd search, and if nothing verifiable turns up, we'd fall back to (i).

Which do you want?

## 11. Learner (RG Sureshkumar)

---

go with option 3

### Agent

---

I'm starting with S7 (Bai et al. 2026), since it analysed token use in agentic coding and may report averages broken down by token type.

[Tool: WebFetch — <https://arxiv.org/html/2604.22750>]

The paper does not provide a comprehensive table breaking down average or median tokens per task by category (input, cached input, output) for each model.

The closest relevant data appears in **Section 5.2 (Phase-Level Token Usage Dynamics)**, which analyzes Claude Sonnet-4.5 only. Figure 8 shows token distributions across phases but presents relative proportions rather than absolute per-task ave ... [truncated]

[Tool: Bash — Search S7 text for token breakdown figures]

et al., 2024 ; Chowdhury et al., 2024 ) . Figure 1 shows the average token usage, monetary costs, and input/output token ratio averaged across all the tasks. On average, agentic coding tasks consume 3500x more tokens than a typical single-round reasoning task and 1200x

more tokens than a multi-round chatting task. Such a gap is primarily driven by the exponential growth of input tokens. Agentic w ... [truncated]

[Tool: ToolSearch — select:WebSearch]

(no output)

[Tool: WebSearch — search: coding agent session token usage breakdown cache read input output tokens per session median study]

[Tool: WebSearch — search: "cache\_read\_input\_tokens" Claude Code average session tokens analysis]

Web search results for query: "coding agent session token usage breakdown cache read input output tokens per session median study"

Links: [{"title": "[BUG] /usage session Breakdown \"Output\" undercounts by ~20x – only counts plain text blocks, drops tool\_use and thinking tokens · Issue #97285 · anthropics/claude-code", "url": "https://github.com/anthropics/claude-code/issues/97285"}, {"title": "Evolv ... [truncated]

Web search results for query: ""cache\_read\_input\_tokens" Claude Code average session tokens analysis"

Links: [{"title": "Cache read tokens consume 99.93% of usage quota – architectural scaling issue with CLAUDE.md re-reads · Issue #24147 · anthropics/claude-code", "url": "https://github.com/anthropics/claude-code/issues/24147"}, {"title": "Where Do Your Claude Code Tokens Actually Go? We Traced Every ... [truncated]

[Tool: WebFetch — https://arxiv.org/abs/2607.13080]

[Tool: WebFetch — https://arxiv.org/abs/2607.02436]

#### # Paper Summary

**\*\*Full Title:\*\*** "Reasoning effort, not tool access, buys first-try reliability in agentic code generation: an observational study"

**\*\*Author:\*\*** Achint Mehta

**\*\*Date:\*\*** Submitted July 2, 2026; revised September 5, 2026

**\*\*Abstract:\*\*** This observational study examined 90 independent agent runs building a real-time retrospective board application. The research evaluated "frontier mod ... [truncated]

#### # Paper Analysis

**\*\*Title:\*\*** Inference Economics of Enterprise Coding Agents: A Case Study of Cloud vs. On-Premise LLMs

**\*\*Authors:\*\*** Sheng-Wei Peng, Yi-Hsun Lin, Yi-Pei Lee

**\*\*Date:\*\*** Submitted 13 July 2026

**\*\*Abstract:\*\*** The study compares API-based Claude Opus versus on-premise GLM-5.1/5.2 models over two 28-day periods on production code, finding prompt caching achieved "99.3% hit rate" and cu ... [truncated]

[Tool: Bash — Search three papers for per-session token breakdowns]

```
=== 2607.02436
=== 2605.09018
.4 Computational Cost All coding-agent sessions were run via Codex using GPT 5.4 at medium reasoning effort. For agent-side usage, we report the cumulative equivalent-token metric T_{eq} (Appendix A.1), which aggregates cached input, fresh input, and output tokens into a single normalized usage scale. This allows runs with different cache and output pr ... [truncated]
```

I found a usable breakdown in an enterprise case study (arXiv:2607.13080). Next I'm checking whether it reports a session count, so its totals can be turned into a per-session example.

[Tool: Bash — Search case study for session counts]

omise low-marginal-cost scaling and data sovereignty at some loss of reasoning fidelity. We study this trade-off through a single-developer, non-randomized longitudinal case study over two contiguous 28-day periods on a production monorepo: an API-based Claude Opus 4.7/4.8 configuration using Claude Code versus an on-premise GLM-5.1/5.2 configuration using Opencode , quantized to NVFP4, on NVIDIA ... [truncated]

I finished the search. One new source has the full breakdown, but it isn't at the session level.

**The candidate:** Peng, Lin & Lee (2026), *Inference Economics of Enterprise Coding Agents: A Case Study of Cloud vs. On-Premise LLMs*, [arXiv:2607.13080](https://arxiv.org/abs/2607.13080). I read the relevant table directly. It covers one developer on a production codebase, using Claude Code with Claude Opus 4.7/4.8, over **28 days**:

| Token type                     | Tokens         |
|--------------------------------|----------------|
| Fresh (incremental) input      | 980,326        |
| Cache creation (writes)        | 108,060,378    |
| Cache reads                    | 15,202,019,495 |
| Output                         | 20,156,852     |
| <b>Total</b> (31,901 requests) | 15,331,217,051 |

**Limitations:** - It's a preprint by a single developer. - It reports no session count, only the 28-day period, and says sessions often ran unattended around the clock. So it doesn't describe a typical session. - It does fit the feature's **project total**, because it's a real 28-day project with every token type broken down.

**What this example shows, using our own arithmetic:** - **Output:** 20.2M tokens × the calculator's Opus 4.8 rate (459 Wh per 100k) is about **93 kWh**. - **Fresh input:** 0.98M tokens at 1–4 Wh per 10k is about **0.1–0.4**

**kWh. - Cache reads:** at S8's assumed rate, 15.2B tokens would be about **590 kWh**, far more than everything else.

So the unknown cached-token figure would dominate any realistic project total. That's the key uncertainty to deal with in a later step.

**Two side findings: - There are four token types, not three.** Both this paper and S8 log cache writes separately. The approved feature lists three types (fresh input, cached input, output), so we'll need to decide where cache writes go. - **A discrepancy with the approved research.** `research.md` says S7 found agentic coding used "about 1,000x" more tokens than code chat or reasoning. S7's text actually says **3,500x** more than single-round reasoning and **1,200x** more than multi-round chat. Research is approved, so whether to correct it is your call.

**Question:** How should we use this?

- **(i) Two separate examples.** The project total fields are pre-filled with this 28-day example, clearly labelled as one developer's around-the-clock use. The session fields start empty, with S8's median session as a reference note, as in option (i) earlier. This requires adding the paper as a new source (S59) to `research.md` for your assessment.
- **(ii) No pre-filled values anywhere.** Both S8 and this case study appear only as labelled reference notes.
- **(iii) Don't use this source**, and fall back to the earlier option (i): empty fields plus the S8 note.

## 12. Learner (RG Sureshkumar)

---

go with option 1

### Agent

---

*[Tool: Bash — Find last source assessment in research]*

```
781:### S55. Budescu, Broomell & Por (2009)
789:### S56. Cain, Loewenstein & Moore (2005)
797:### S57. Mu & Lee (2023)
805:### S58. Ma et al. (2024)
813:## Selected features
S58. Ma et al. (2024)
```

- Citation: Ma, Y., Zhao, S., Chen, J., Hu, C., & Qu, J. (2024). Perceived greenwashing and employee green behavior: The roles of green organizational identity and self-serving leadership. \*Business ... [truncated]

*[Tool: Edit — research.md]*

*[Tool: Edit — spec.md]*

*[Tool: Bash — Find insertion point for feature 2]*

The file research.md has been updated successfully.

The file spec.md has been updated successfully.

83:## User approval

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've added S59 to research.md as a new source, marked "not yet assessed by the user", with its limitations and a proposed decision for you to accept or change. I've also started a Feature 2 draft in [spec.md](#) with your decisions recorded and three open items listed.

Separately, the S7 "1,000x" discrepancy in the approved research is still there. I haven't touched it. Tell me if you want it corrected.

### Open item: cache-write tokens

Real tools log four token types: fresh input, cache writes, cache reads and output. The approved feature has three. Here's what the evidence says about cache writes: - **What they are:** new context being processed for the first time and then stored. S7 observes that fresh-input and cache-write tokens "track each other closely", because new context is cached as soon as it enters. - **Energy:** no source measures them separately. S8's only figure is 1.25x fresh input, taken from API prices.

**Question:** How should cache writes be handled?

- **(a) Four input fields.** Cache writes get their own field and are costed at the fresh-input rate (1–4 Wh per 10,000 tokens). The interface explains that the processing work is the same as fresh input. This matches what tools actually log, so employees can copy numbers across directly, but it adds a field that isn't in the approved three.
- **(b) Three input fields.** Employees are told to add cache writes into "fresh input" themselves. This keeps the approved wording, but relies on employees doing the addition correctly.

Both options give the same energy result. The difference is whether the tool or the employee does the addition. Which do you want?

## 13. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: Bash — Record cache-write decision and revision note]

[Tool: Bash — Show current Revisions section]

## ## Revisions

If implementation changes the intended result, update the specification and record what changed and why.

- 2026-09-26 (specification stage): Feature 1 as approved in `research.md` logged images "by model type and settings (resolution, quality or steps)". The user decided to use one general range with no model or settings inputs, because no commercial tool has published figures, the ... [truncated]

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded the four-field decision and added a Revisions note.

### Open item: energy for cache-read tokens

This is the hardest number in the feature, and it drives the totals: - **No source measures it.** The only figure is S8's 0.39 Wh per 10,000 tokens. That's one-tenth of S8's fresh-input rate, taken from API prices, and its author doesn't know whether it's reasonable. - **What the mechanism tells us.** A cache read skips reprocessing the input, but it isn't free: the stored context is still used when the model generates, and storage has its own hardware footprint (S12, S10). - **The bounds would be ours, not a source's.** We can reason that a cache read should cost somewhere between nearly zero and the fresh-input rate. That's reasoning rather than a sourced figure, and it would have to be labelled that way.

What each option means in practice, using our own arithmetic:

| Option                                   | Cached-token rate<br>(per 10,000 tokens) | S8-scale session (~570k<br>cache reads)     | S59 28-day project (15.2B cache reads)                                              |
|------------------------------------------|------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------|
| (a) Reasoned bounds,<br>S8 as central    | 0 to 4 Wh, central 0.39<br>Wh            | 0–228 Wh, central about<br>22 Wh            | 0–6,100 kWh, central about 590 kWh                                                  |
| (b) Cache reads kept<br>out of the total | none                                     | shown separately as an<br>unquantified item | shown separately; the project total covers<br>only about 100–140 kWh of known parts |

- **(a)** keeps cache reads in the total, but with a very wide range, and its central value rests on a price-based assumption. It would be explicitly labelled as "our estimate, no measurement exists".
- **(b)** keeps unsupported numbers out of the total. But the total then leaves out what is probably the largest component, and a reader could mistake it for the whole footprint.

**Question:** Which approach do you want for cache reads, or would you rather handle it some other way?

## 14. Learner (RG Sureshkumar)

go with option a

## Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded that decision. Cache reads are included in the total at 0–4 Wh per 10,000 tokens, with S8's 0.39 as the central value, and they're labelled "our estimate; no measurement exists".

### Open item: the central value for fresh input (and cache writes)

The range is settled at 1–4 Wh per 10,000 tokens. The AI total also needs a central value, and there are three sourced candidates:

- **(a) About 2.1 Wh per 10,000 tokens (S10, Epoch AI).** This is a compute model for GPT-4o, rated moderate confidence, and it sits in the middle of the range.
- **(b) About 1.7 Wh per 10,000 tokens (S11).** This is inferred from API latency and rated moderate to low confidence. The research treats it as an upper estimate for that method.
- **(c) About 3.9 Wh per 10,000 tokens (S8).** This is a price-based proxy, rated low confidence. It is the figure S8's 0.39 cache-read value was derived from, so the cache-read central would be exactly one-tenth of it.

If you choose (a) or (b), the cache-read central (0.39) would be about one-fifth or one-quarter of fresh input rather than one-tenth. I'd note that in the methodology rather than change the decision you just made.

For output tokens, I'll propose a default for you to object to if you disagree. The session gets a model choice from the calculator's existing list, and output energy comes from that model's EcoLogits figure. The input rates are the same for every model, because no source gives them per model, and the interface would say so.

**Question:** Which central value should fresh input use?

## 15. Learner (RG Sureshkumar)

---

go with option a

## Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded 2.1 Wh per 10,000 tokens as the central value, with a methodology note that the cache-read central is therefore about one-fifth of fresh input. I've also recorded the model-choice default for output tokens, noted as my proposal that you didn't object to.

## Open item: where sessions are entered, and what happens to the old "agent session"

The output-only "coding / agent session" (100,000 output tokens) currently appears in two places: - as a size in the calculator's typed-in rows - as the tracker's "Agent session" quick-log button

The approved feature says the new estimator **replaces** that flawed figure.

**Question:** Where should employees enter agent sessions?

- **(a) Tracker only, matching the image decision.** The tracker's "Agent session" button opens the token-entry form (model plus four token fields). The old output-only size is removed from both the tracker and the calculator. This is consistent with feature 1 and removes the flawed figure completely. As with images, sessions wouldn't count when "Use my log" is off.
- **(b) Both places.** The tracker gets the form, and the calculator's agent size is also replaced with token entry. Sessions count in both modes, but the work and the interface are heavier.
- **(c) Tracker form only, but the old size stays in the calculator** with a warning that it counts output tokens only. This is the smallest change, but the known flaw stays visible in part of the tool.

We'll deal with how project totals relate to daily totals, and how to avoid counting the same sessions twice, in the next step. Which do you want?

## 16. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded that: sessions are logged in the tracker only, and the old output-only size is removed everywhere.

## Open item: how project totals relate to the daily AI total

There are two ways a project total could come about, and they could count the same sessions twice: 1.

**Summing logged sessions.** Sessions logged day by day in the tracker, which already count toward each day's AI total. 2. **Direct entry.** The project fields you chose to pre-fill with S59's 28-day example, where someone types in a whole project's token counts at once.

**Question:** How should project totals work?

- **(a) Both, kept apart.** When logging a session, the employee can optionally attach a project name. A project view sums a project's tagged sessions across days, and those sessions count once, on the day they were logged. Separately, a "project estimate" form (pre-filled with S59) calculates a whole project

from typed-in totals, but it is shown **beside** the AI total and never added to it, so nothing is counted twice. Jordan gets both a tracked project and a what-if estimate, at the cost of more interface.

- **(b) Direct entry only.** The project form (pre-filled with S59) is a standalone estimate shown beside the AI total and not added to it. Logged sessions count toward daily totals but aren't grouped into projects. This is simpler, but the project total is never connected to what was actually logged.
- **(c) Tagged sessions only.** The project total is always the sum of logged sessions. This removes the direct-entry form and its S59 pre-fill, reversing your earlier decision, and S59 would then appear only as a reference note.

Which do you want?

## 17. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded that: tagged sessions add up into a project view, and the S59-pre-filled project estimate sits beside the AI total without being added to it.

### Open item: carbon and water for input tokens

Output tokens already come with EcoLogits' carbon (including embodied hardware carbon) and water, per model. Input tokens only have energy figures, so their carbon and water need a method. Here's what I propose, following the approach you chose for images:

- **Carbon:** input energy × the selected region's grid intensity, the same as every other electricity figure. Embodied hardware carbon is counted only through EcoLogits' output figure, because no source gives one for input tokens. The interface would say so.
- **Water:** input energy × **4.88 L/kWh** (S46), labelled as derived, the same factor as images. Output water stays as EcoLogits reports it.

The catch is that one session's water would combine two factors: EcoLogits' own intensity for output (roughly 3.6–4.1 L/kWh, implied by its figures) and S46's 4.88 for input. The difference is small compared with the ranges, but it would be disclosed.

The alternative would be to derive the input water factor from EcoLogits' implied ratio for the chosen model. That's more internally consistent, but it's a back-calculated number rather than a published factor, which is why you turned down that route for images.

**Question:** Should input tokens follow the image method (grid carbon, S46 water at 4.88 L/kWh, embodied carbon on output only), or do you want something different?

## 18. Learner (RG Sureshkumar)

---

follow the image method

### Agent

---

[Tool: Bash — Compute worked acceptance-check figures]

```
out [22.9521, 15.685585, 30.218635] in [4.2, 2.0, 8.0] cache [19.5, 0.0, 200.0]
E [46.652100000000004, 17.685585, 238.218635] C [18.178213, 7.1709373, 90.97349630000001] W
mL [201.720185, 60.75955, 1136.168815]
out [92528.41655784, 63234.403075684, 121822.510667404] in [22898.547840000003, 10904.0704,
43616.2816] cache [592878.760305, 0.0, 6080807.798]
E [708305.72470284, 74138.473475684, 6246246. ... [truncated]
```

[Tool: Edit — spec.md]

[Tool: Bash — Remove open-items note from feature 2 and check none remain]

```
The file spec.md has been updated successfully.
```

```
156>User: Open the project repository as your workspace, start a fresh chat, and type
`start specification`.
```

Feature 2 is now fully drafted in [spec.md](#), under **Features** → **Feature 2**, with no open items left.

**How a session is calculated:** - **Output:** tokens × the chosen model's EcoLogits rate. - **Fresh input and cache writes:** 2.1 Wh per 10,000 tokens (range 1–4). - **Cache reads:** 0.39 Wh per 10,000 tokens (range 0–4), labelled as our own estimate. - **Carbon:** energy × the grid intensity, plus embodied carbon on output only. - **Water:** EcoLogits' figure for output, plus input energy × 4.88 L/kWh.

**Worked checks, all calculated from the decided figures:** - **One session** on Claude Opus 4.8 in the US region (10k fresh, 10k cache writes, 500k cache reads, 5k output tokens): **about 47 Wh (18–238 Wh), 18 g CO<sub>2</sub>e and 202 mL**. Cache reads alone account for 0–200 Wh of that range, which shows how much the uncertain component drives the result. - **The S59 28-day project estimate: about 708 kWh (74–6,246 kWh), 271 kg CO<sub>2</sub>e and 3,350 L**. It's shown beside the AI total and never added to it.

Some items are my proposals rather than your decisions, so please check these: - **Breakdown by component.** Each logged session shows how its total splits across output, fresh input and cache writes, and cache reads. - **"Undo last" and the budget.** Sessions can be undone like prompts and count toward the monthly budget when the log switch is on. - **Two new limitations in the interface.** It would say that a flat per-token rate ignores the fact that input cost grows faster than context length (S10). It would also say that the same task can vary up to 30× in tokens (S7).

**Question:** Does the Feature 2 draft look right, or is there anything to change before we move to feature 3, the digital day builder?

## 19. Learner (RG Sureshkumar)

---

its good, lets move to feature 3

### Agent

---

#### Feature 3: digital day builder

The approved feature lets employees enter hours of video calls, video and music streaming, social media and gaming by device (phone, laptop, TV, console, PC). The result goes into the separate **digital-day total**.

The first decision is the accounting boundary, meaning which parts of an hour get counted. The research shows the choice changes results several-fold. S25's breakdown of one hour of European video streaming in 2020 is:

| Part           | Wh per hour | Share |
|----------------|-------------|-------|
| Viewing device | 96          | 51%   |
| Home router    | 71          | 38%   |
| Network        | 20          | 10%   |
| Data centres   | about 1     | 1%    |

**A like-for-like issue that bears on feature 5:** the AI figures (EcoLogits, images, sessions) count data centres only. They don't count the laptop or phone the employee uses to prompt. If the digital-day total includes devices and the AI total doesn't, the two totals side by side aren't measured the same way.

**Question:** What should an hour of digital activity count?

- **(a) Device, network and data centres, but not the home router.** The router is usually on anyway, and S25 notes its share depends on how shared energy is allocated. This captures the dominant device share without counting always-on infrastructure. The interface would state plainly that the AI total excludes the employee's own device.
- **(b) Everything, including the router.** This follows S25's full figure, but it counts a router that runs regardless of the activity, and it would be the least comparable with the AI total.
- **(c) Network and data centres only, excluding the device.** This matches the AI total's boundary exactly, but it leaves out what every source identifies as the largest part, so the digital-day totals would be small and would hide the device differences (TV against phone) that the feature exists to show.

Which do you want?

## 20. Learner (RG Sureshkumar)

---

go with option b

### Agent

---

*[Tool: Bash — Read source assessments S18–S28]*

```
S18. Obringer et al. (2021)
```

```
- Citation: Obringer, R., Rachunok, B., Maia-Silva, D., Arbabzadeh, M., Nateghi, R., &
Madani, K. (2021). The overlooked environmental footprint of increasing Internet use.
Resources, Conservation and Recycling, 167, 105389.
<https://doi.org/10.1016/j.resconrec.2020.105389>
- Claim or figure: one hour of videoconferencing or streaming emits 150–1,000 g CO2 and re
... [truncated]
```

*[Tool: Edit — spec.md]*

*[Tool: Bash — Download S25 white paper and search device power figures]*

The file spec.md has been updated successfully.

```
Exit code 1
ct.pdf: PDF document, version 1.7
Traceback (most recent call last):
 File "<string>", line 1, in <module>
ModuleNotFoundError: No module named 'pypdf'
```

*[Tool: Bash — Extract S25 text and search device figures]*

```
You should consider upgrading via the '/Library/Developer/CommandLineTools/usr/bin/python3
-m pip install --upgrade pip' command.
fontTools is required to fully parse the encoding of a CFF Type1 font in font dictionary
{'/BaseFont': '/CEJKOJ+MyriadPro-Regular', '/Encoding': '/WinAnsiEncoding', '/FirstChar':
67, '/FontDescriptor': IndirectObject(712, 0, 4311842768), '/LastChar': 82, '/Subtype': '/T
... [truncated]
```

*[Tool: Bash — Read S25 device and router assumptions]*

```
Transmission – Network
(Core & Access)
Fixed network
• Energy intensity, IFN (2020) = 0.0065kWh/GB
• Derived from Aslan et. al, 2018 using the regression analysis presented in the paper
Mobile network
• Energy intensity, IMN (2020) = 0.1kWh/GB
• Sourced from Pihkola et. al, 2018
Transmission – Home router
• Energy intensity, IHR (2019) = 0.025kWh/GB
• Based on 10W home router (only used with fi ... [truncated]
```

*[Tool: Bash — Locate S25 appendix device power table]*

```
163:10. Appendix 99
1876:Ws = average power consumption of screens
```

1877:Wp = average power consumption of peripherals  
1954:• Reasonable estimates of average power (W) for specific devices, see Appendix for details  
2056:see Appendix for detailed information. The energy consumption  
2071:the average power consumption of screens, Wp is the average  
2085:in the Appendix.  
2115:Appendix), and the modell ... [truncated]

I'm reading the appendix of the Carbon Trust streaming paper (S25), which lists the device power figures we'll need for feature 3.

*[Tool: Bash — Read S25 device power appendix]*

101  
Carbon impact of video streaming  
Appendix  
Table 10. Estimated power consumption of devices for streaming  
Device type Power Draw (W) Comment Source  
Android Phone 1  
Samsung Galaxy S9, has 11.55Wh  
battery with up to 16 hours lifetime when  
watching videos  
iFixit for battery Wh, Samsung for  
battery lifetime  
iPhone 1 Assuming ~11Wh battery and 10h of  
battery life if streaming Apple, 2021a, Ap ... [truncated]

I read S25's methodology and appendix directly. They show something that affects the router decision you just made.

**The 71 Wh router figure isn't the router's actual electricity.** S25's "conventional approach" assigns router energy by data volume: 0.025 kWh per GB, based on a **10 W** router and average household data use. An hour of streaming moves about 3 GB, so the router gets charged about 71 Wh. But the paper's own comparison table lists a home router running for one hour at **10 Wh**. The 71 Wh is therefore an allocation by data volume, which is exactly the per-gigabyte method that S19 (peer-reviewed) says doesn't reflect how network equipment actually uses energy. S25 also notes the router applies only to fixed home broadband, not phones on cellular.

The same appendix (S25 Table 10) gives the device power figures we'll need next:

| Device                        | Power (W) | Note                                         |
|-------------------------------|-----------|----------------------------------------------|
| Smartphone                    | 1         | from battery capacity and video battery life |
| Laptop                        | 22        |                                              |
| Desktop computer with monitor | 115       |                                              |
| TV (smart or normal)          | 100       | "conservative estimate for 2020"             |
| Gaming console (PS4)          | 89        |                                              |
| Set-top box                   | 18        |                                              |

**Question:** You chose to count everything, including the router. How should the router be counted?

- **(a) Its actual draw: 10 Wh per hour** on home Wi-Fi (S25's 10 W figure), and nothing on cellular. This still counts the router, as you decided, but uses its physical power rather than a per-gigabyte allocation.
- **(b) S25's allocation of 71 Wh per hour** for video on home broadband, scaled by data volume for other activities. This is S25's headline method, but it's the per-gigabyte approach that S19 criticises, and the router would appear to use 7× its rated power.

Which do you want?

## 21. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

*[Tool: Bash — Locate S25 power-model results]*

```
132:5.2. Malmodyn's simple power model provides a closer representation of the dynamics of
internet transmission 55
133:5.3. Methodology of the power model approach 57
134:5.4. Overview of the power model approach 57
138:5.8. Further validation and refinement of the power model is a logical next step 66
210:a power model approach, which uses a marginal allocation
213:data volume used. The power m ... [truncated]
```

*[Tool: Bash — Read S25 power-model assumptions and results]*

```
carbon impact of video streaming.
The aim is to contribute to the understanding of the topic,
so that future decisions can be based on an informed
understanding of the issue, with an insight of the methods,
uncertainties and variability that can affect the estimates.
The first is the conventional approach, which is well
established and has been used in most previous studies, and
follows an ... [truncated]
```

*[Tool: Edit — spec.md]*

```
The file spec.md has been updated successfully.
```

I've recorded the router decision: 10 Wh per hour on home broadband or Wi-Fi, and none on cellular. I also added one small point marked **Open**, which is my suggestion and not something you've decided: don't count the router twice when two activities overlap in the same hour. Tell me whether you want it.

**The same issue affects the network figure.** S25's 20 Wh per hour for the network also comes from a per-gigabyte rate (0.0065 kWh per GB on fixed broadband, and 0.1 kWh per GB on mobile). S25 also ran a **power model**, based on Malmodyn's work, which S19 supports. That model found network energy barely

changes with data volume: - **Fixed broadband:** "just under 1 gCO<sub>2</sub>e/hour to just over 1 gCO<sub>2</sub>e/hour" from SD to 4K. - **Mobile (4G):** about **0.5 g** on the data-saver setting, rising to nearly **3.5 g** at maximum.

S25 reports these in grams on the European grid, not in Wh. Converting at S25's own ratio (56 g per 188 Wh, about 0.3 g per Wh) gives:

| Connection      | Wh per hour (derived by us)           |
|-----------------|---------------------------------------|
| Fixed broadband | about 3–4                             |
| Mobile          | about 2–12, depending on data setting |

These conversions are our own derivation and would be labelled as such.

**Question:** Which method should the network use?

- **(a) The power model.** About 3–4 Wh per hour on fixed broadband and about 2–12 Wh on mobile, labelled as derived from S25. This is consistent with your router decision and with S19, and the network stays a small share.
- **(b) The per-gigabyte rates.** 0.0065 kWh per GB fixed and 0.1 kWh per GB mobile, times each activity's data use (for example 0.5–3.4 GB per hour for video calls, S22). This is S25's headline method, but it's the approach S19 criticises, and it's inconsistent with how you've handled the router.

Which do you want?

## 22. Learner (RG Sureshkumar)

go with option b

### Agent

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded the per-gigabyte network method, with a disclosure that S19 disputes it and that the router and network are counted by different methods.

Under this method, every activity needs a data-use figure per hour. The research covers some activities but not others:

| Activity        | Data per hour                                         | Source                                |
|-----------------|-------------------------------------------------------|---------------------------------------|
| Video streaming | SD 1 GB, HD 3 GB, 4K 7 GB (fixed); 0.17–3 GB (mobile) | S25, from Netflix's published figures |
| Video calls     | 0.5–3.4 GB, depending on the app                      | S22                                   |
| Music streaming | <b>none verified</b>                                  | none                                  |

| Activity     | Data per hour        | Source                                                                         |
|--------------|----------------------|--------------------------------------------------------------------------------|
| Social media | <b>none verified</b> | S29 and S30 were rejected, and S32 (Meta) is per person per year, not per hour |
| Gaming       | <b>none verified</b> | local play uses little data; cloud gaming wasn't quantified                    |

**Question:** How should the three activities without a data-use figure be handled?

- **(a) Search for sources now.** For example, streaming services' own published data use per quality setting, or platform or measurement figures for social media and online gaming. Anything found would be added to `research.md` for your assessment, and anything I can't verify would fall back to (b).
- **(b) Count the device and router only.** The network share for those activities is left out, and the interface states that no verified figure exists. This is honest about the gap, but those activities would be undercounted relative to video.
- **(c) Use a stand-in.** For example, social media treated as SD video. This is simpler, but it's our own construction, not a sourced figure.

Which do you want?

## 23. Learner (RG Sureshkumar)

go with option a

### Agent

I'm looking for published data-use figures, starting with music streaming.

[Tool: WebFetch — <https://support.spotify.com/us/article/data-saver-mode/>]

**\*\*Page Title:\*\*** "Page not found – Spotify"

**\*\*Data Usage and Bitrate Information:\*\*** The webpage contains no figures about data usage per hour or bitrate specifications for any audio quality settings (low, normal, high, very high, or lossless). This is a 404 error page that only provides navigation options and does not include technical audio specifications.

[Tool: WebSearch — search: Spotify support audio quality settings kbps low normal high very high]

Web search results for query: "Spotify support audio quality settings kbps low normal high very high"

Links: [{"title": "Audio quality – Spotify", "url": "https://support.spotify.com/us/article/audio-quality/"}, {"title": "Lossless Audio Quality – Spotify", "url": "https://support.spotify.com/us/article/lossless-audio-quality/"}, {"title": "Equalizer – Spotify", "url": "https://support.spotify.com/us/article/..."}]  
[truncated]

[Tool: WebFetch — <https://support.spotify.com/us/article/audio-quality/>]

# Audio Quality – Spotify

**\*\*Page Title:\*\*** Audio quality – Spotify

**##** Music Quality Bitrates

For **\*\*Spotify Premium\*\*** on desktop, mobile, and tablet:

- **\*\*Low:\*\*** "Equivalent to approximately 24kbit/s"
- **\*\*Normal:\*\*** "Equivalent to approximately 96kbit/s"
- **\*\*High:\*\*** "Equivalent to approximately 160kbit/s"
- **\*\*Very high:\*\*** "Equivalent to approximately 320kbit/s"
- **\*\*Lossless:\*\*** "Equivalent up to 24 ... [truncated]"

Spotify's support page gives bitrates, which convert to data per hour. Next I'm searching for social media and gaming figures.

*[Tool: WebSearch — search: measurement study data consumption per hour TikTok Instagram Facebook scrolling MB per minute]*

*[Tool: WebSearch — search: online multiplayer gaming data usage per hour MB measured study console PC]*

Web search results for query: "online multiplayer gaming data usage per hour MB measured study console PC"

Links: [{"title": "How much data does gaming use? – Holafly", "url": "https://esim.holafly.com/internet/how-much-data-does-gaming-use/"}, {"title": "How Much Data Does Gaming Use on Mobile, PC & Console", "url": "https://www.jetpacglobal.com/blog/how-much-data-does-gaming-use/"}, {"title": "How Much ... [truncated]"}]

Web search results for query: "measurement study data consumption per hour TikTok Instagram Facebook scrolling MB per minute"

Links: [{"title": "How Much Data Does Facebook Use?", "url": "https://roamless.com/blog/how-much-data-does-facebook-use/"}, {"title": "Data usage calculator | Find out how much data you use | Saily", "url": "https://saily.com/data-usage-calculator/"}, {"title": "What consumes the mo ... [truncated]"}]

*[Tool: WebFetch — https://www.meter.net/news/what-consumes-the-most-data-on-mobile-youtube-spotify-netflix-and-other-apps-under-scrutiny/]*

*[Tool: WebFetch — https://www.microsoft.com/en-us/edge/learning-center/how-much-data-does-gaming-use]*

# Publication Details

**\*\*Publisher & Date:\*\*** The article appears on Meter.net but doesn't specify an author or publication date.

**\*\*Methodology:\*\*** The article lacks disclosure of measurement methods, devices used, or test

duration. No source citations or research methodology is provided.

#### # Social Media Data Usage Figures

The article provides these consumption estimates:

– **Instagram:** "approx ... [truncated]

#### # Data Usage Figures from Microsoft Edge Learning Center Article

**Publication Date:** October 10, 2023

**Author:** Not specified

**Data Usage Per Hour:**

The article provides only one specific hourly figure:

"some of the most popular online games consume around 100 Megabytes (MB) of data per hour"

This figure applies to online games generally but does not distinguish between console, PC, or ... [truncated]

*[Tool: WebSearch — search: traffic characterization online game Fortnite bandwidth kbps measurement paper arXiv]*

*[Tool: WebSearch — search: TikTok short video traffic measurement study data volume per session arXiv]*

Web search results for query: "traffic characterization online game Fortnite bandwidth kbps measurement paper arXiv"

Links: [{"title": "(PDF) Measurement and characterization of Internet gaming traffic", "url": "https://www.academia.edu/7465676/Measurement\_and\_characterization\_of\_Internet\_gaming\_traffic"}, {"title": "Modelling Dead Rocking In Online Multiplayer Games", "url": "https://arxiv.org/pdf/1208 ... [truncated]"}]

Web search results for query: "TikTok short video traffic measurement study data volume per session arXiv"

Links: [{"title": "Analyzing User Engagement with TikTok's Short Format Video Recommendations using Data Donations", "url": "https://arxiv.org/html/2301.04945v2"}, {"title": "Analyzing User Engagement with TikTok's Short Format Video", "url": "https://arxiv.org/pdf/2301.04945"}, {"title": "Short-Form ... [truncated]"}]

*[Tool: Bash — Download Know Thy Lag paper and search bandwidth figures]*

head: lag.txt: No such file or directory  
ugrep: warning: lag.txt: No such file or directory  
Shell cwd was reset to .

[Tool: Bash — Check download status of the paper]

```
ls: lag.pdf: No such file or directory
```

I've finished the search. Here's what turned up:

**Music streaming: found and verified.** Spotify's own [Audio quality](#) support page gives bitrates for each setting: Low about 24 kbit/s, Normal about 96, High about 160 and Very high about 320. Those convert to about **11, 43, 72 and 144 MB per hour** (the conversion is my arithmetic). This is company documentation of its own settings, which is appropriate for a technical fact like this. I'd add it to `research.md` as S60 for you to assess.

**Social media: nothing verifiable.** - Every per-hour figure I found (for example "TikTok about 840 MB per hour" or "Instagram about 720 MB per hour") comes from eSIM sellers or sites like [meter.net](#) that don't say how they measured. - The academic TikTok papers I found study viewing behaviour, not data volume.

**Gaming: nothing verifiable.** - [Microsoft's Edge Learning Center](#) says popular online games use "around 100 MB" per hour, but it cites commercial blogs. - The research literature on game traffic suggests a few hundred kbit/s at most, but I couldn't retrieve a paper to check the figures.

Under the rule you set, social media and gaming therefore fall back to option (b): device and router only, with the missing network share disclosed. **How much that gap matters depends on the activity:** - **Gaming:** even 100 MB per hour on broadband is under 1 Wh, against about 216 W for a PS5. The gap is negligible. - **Social media on cellular:** if the unverified 0.1–0.84 GB per hour were right, the network share would be about **10–84 Wh per hour** at 0.1 kWh per GB, which could far exceed a phone's 1 W. Leaving it out could noticeably understate social media on mobile, and the interface would need to say so plainly.

**Question:** Should I go ahead as described, adding Spotify as S60 and falling back to device and router only for social media and gaming, or do you want to handle social media differently given how large the cellular gap could be?

## 24. Learner (RG Sureshkumar)

---

go ahead as described

### Agent

---

[Tool: Edit — `research.md`]

[Tool: Edit — `spec.md`]

```
The file research.md has been updated successfully.
```

```
The file spec.md has been updated successfully.
```

I've added Spotify to `research.md` as S60, marked "not yet assessed", with a note recording the unsuccessful search for social media and gaming. In the `spec`, the data-use figures are recorded along with the plain-

language disclosure about the social media gap on cellular.

**Next: device power.** Here is a proposed figure for each device, using only sources already in the research:

| Device                      | Proposed power | Source                                   | Caveat                                               |
|-----------------------------|----------------|------------------------------------------|------------------------------------------------------|
| Smartphone                  | 1 W            | S25                                      | derived from battery capacity and video battery life |
| Laptop                      | 22 W           | S25                                      |                                                      |
| Office desktop with monitor | 115 W          | S25                                      |                                                      |
| TV                          | 100 W          | S25                                      | S25 calls it "conservative for 2020"                 |
| Console, gaming             | 214–219 W      | S33 (Sony, PS5, 2026)                    | S25's 89 W is for an older PS4                       |
| Console, streaming or menus | 43–47 W        | S33 (PS5 home menu)                      | a stand-in; Sony gives no streaming-app figure       |
| Gaming PC, gaming           | 305–359 W      | S37 (modelled) to S36 (measured, one PC) | <b>excludes the monitor</b>                          |

**The main gap is the gaming PC's monitor.** No source in the research gives a monitor's power separately. S25's 115 W includes a monitor but doesn't break it out. S35 includes the display but uses 2015 hardware and gives yearly totals.

**Question:** How should the gaming PC's monitor be handled?

- **(a) Leave it out.** Use 305–359 W for the PC alone, and state plainly that the monitor isn't counted. This is honest, but it undercounts Jordan's gaming.
- **(b) Search for a sourced monitor figure** (for example ENERGY STAR data or a manufacturer's specification), added to `research.md` for your assessment. If nothing verifiable turns up, we'd fall back to (a).

The rest of the table is also open to your changes if any figure looks wrong to you.

## 25. Learner (RG Sureshkumar)

go with option b

### Agent

I'm looking for an authoritative monitor power figure, starting with ENERGY STAR's certified-displays data.

[Tool: WebSearch — search: ENERGY STAR certified displays dataset on mode power watts data.energystar.gov monitors]

Web search results for query: "ENERGY STAR certified displays dataset on mode power watts data.energystar.gov monitors"

```
Links: [{"title":"Product Finder – ENERGY STAR Certified Displays","url":"https://www.energystar.gov/productfinder/product/certified-displays/details/2352049"}, {"title":"Draft 1 Version 8.0 ENERGY STAR Displays Specification","url":"https://www.energystar.gov/sites/default/files ... [truncated]}
```

*[Tool: Bash — Download ENERGY STAR displays dataset and list columns]*

```
2891 displays.csv
1 ENERGY STAR Unique ID
2 ENERGY STAR Partner
3 Brand Name
4 Model Name
5 Model Number
6 Additional Model Information
7 UPC
8 Product Type
9 Panel Type
10 Native Resolution (pixels)
11 Total Native Resolution (megapixels)
12 Screen Size (inches)
13 Screen Area (square inches)
14 Model Features
15 Features Enable ... [truncated]
```

*[Tool: Bash — Summarise monitor on-mode power by screen size]*

```
Counter({'Monitor': 1592, 'Signage Display': 1296})
21 25 644 median 12.565000000000001 p10 10.4 p90 14.81
26 28 532 median 16.92 p10 12.49 p90 25.2
29 35 217 median 28.19 p10 20.03 p90 34.07
0 100 1588 median 14.17 p10 10.0 p90 29.02
01/02/2024 12:00:00 AM 12/31/2021 12:00:00 AM
Shell cwd was reset to .
```

*[Tool: Bash — Check certification date range of monitors]*

```
2016-06-23 2026-09-23
Shell cwd was reset to .
```

I found a verifiable source and summarised it myself. It's the EPA's **ENERGY STAR Certified Displays** dataset ([data.energystar.gov](https://data.energystar.gov)), downloaded today. It lists 1,592 certified monitors, certified between 2016 and September 2026, each with its measured on-mode power.

My summary of on-mode power:

| Screen size | Monitors | Median | Middle 80% (P10–P90) |
|-------------|----------|--------|----------------------|
| 21–25 in    | 644      | 12.6 W | 10.4–14.8 W          |
| 26–28 in    | 532      | 16.9 W | 12.5–25.2 W          |
| 29–35 in    | 217      | 28.2 W | 20.0–34.1 W          |

| Screen size  | Monitors | Median | Middle 80% (P10–P90) |
|--------------|----------|--------|----------------------|
| All monitors | 1,588    | 14.2 W | 10.0–29.0 W          |

**Limitations:** - **Certified models only.** These are likely more efficient than the average monitor in use. - **Test conditions.** Power is measured at default brightness with a standard test signal, not during gaming. - **Gaming monitors not separated.** The dataset doesn't identify high-refresh-rate gaming monitors, which may draw more.

**Proposal:** add **14 W (range 10–29 W)** for the monitor to the gaming PC's 305–359 W. That gives a gaming PC total of about **319–388 W**. I'd add the dataset to `research.md` as S61 for you to assess, with the summary statistics marked as my own calculation from the raw data.

**Question:** Do you want to use the all-monitor figure (14 W, 10–29 W), or the 26–28 inch figure (16.9 W, 12.5–25.2 W), since many gaming monitors are that size? The size-based choice would be our assumption, not a sourced fact.

## 26. Learner (RG Sureshkumar)

---

use all the monitors

### Agent

---

[Tool: Edit — `research.md`]

[Tool: Edit — `spec.md`]

The file `research.md` has been updated successfully.

The file `spec.md` has been updated successfully.

I've added the ENERGY STAR dataset to `research.md` as S61 (not yet assessed), and the full device table is now in the spec.

**Next: how employees enter digital activities.** The page currently works in two modes. There's the calculator's typed-in "average day", and there's the tracker's live log, which takes over the page when "Use my log" is on. Images and agent sessions are tracker-only. Digital activities are a different kind of input: hours of streaming or calls are hard to log live, one at a time.

**Question:** How should employees enter their digital day?

- **(a) A separate "typical day" panel.** Each row is an activity, a device, a connection (home Wi-Fi or cellular), hours per day, and a quality setting where it applies (video resolution or music quality). The digital-day total is calculated from this panel, whatever the log switch is set to, because it's a separate total from AI anyway.
- **(b) Tracker only.** Employees log each call, stream or gaming session as it happens, as with images and agent sessions. This is consistent with features 1 and 2, but awkward for hours of background

activity, and the digital-day total would depend on the log switch.

- **(c) Both.** A typical-day panel plus optional tracker logging. This is the most flexible option, but it's the most interface work, and we'd need a rule for which one counts when both are filled in.

Which do you want?

## 27. Learner (RG Sureshkumar)

---

go with option c

### Agent

---

Both entry routes it is. That needs a rule for when an employee has filled in both.

**Question:** When both the typical-day panel and tracker logs exist, which counts toward the digital-day total?

- **(i) Follow the existing log switch.** With "Use my log to drive this page" on, the digital-day total uses today's logged activities. With it off, it uses the typical-day panel. This matches how the AI total already behaves, but early in the day the logged total will be small and look incomplete.
- **(ii) Override activity by activity.** An activity that has been logged today replaces that activity's typical-day entry, and every other activity keeps its typical-day hours. This gives the most complete picture, but it's more complex to build and explain, and it mixes an actual record with an estimate in one total.
- **(iii) Always use the panel; logs are a record only.** Logged activities are shown as a record but never enter the total. This is simple, but it largely defeats the purpose of logging.

Which do you want?

## 28. Learner (RG Sureshkumar)

---

go with option 2

### Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded the activity-by-activity rule. I interpreted "activity" as the activity type. For example, if any video streaming is logged today, it replaces the whole typical-day video streaming entry, whatever device it was on. Tell me if you meant it more finely, such as per device.

**Next: carbon and water for home and network electricity.**

- **Carbon:** I propose energy × the selected region's grid intensity, the same method used everywhere else in the calculator.

- **Water:** the image and session factor (4.88 L/kWh) is specific to data centres, because it includes cooling water. Home devices, routers and networks have no cooling tower, so their water is only what power plants use to generate the electricity. S46 gives that figure for US electricity overall: **4.35 L/kWh**.

**Question:** For digital-day water, do you want:

- **(a) 4.35 L/kWh** (S46, US electricity overall) for device, router and network electricity, and **4.88 L/kWh** for the small data-centre share. This matches each part's actual water scope, but uses two factors.
- **(b) 4.88 L/kWh for everything.** Simpler, but it adds data-centre cooling water that home devices don't use. The difference is only about 12%.

Both factors are US averages, so the interface would note that water doesn't change with the region setting, unlike carbon.

## 29. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded water at 4.35 L/kWh for home and network electricity, and 4.88 L/kWh for data centres.

The router question from earlier is still open. Take someone who streams on the TV while scrolling social media on the phone, both on home Wi-Fi, for the same hour. If we count the router's 10 Wh once per activity, that hour gets 20 Wh of router energy, although a single router only draws 10 W.

**Question:** How should overlapping activities on home Wi-Fi be counted?

- **(a) Per activity.** Each activity-hour gets its own 10 Wh. This is simple, but overlapping use overcounts the router.
- **(b) Capped at the day's connected hours.** The router is counted for the total hours of Wi-Fi activity, up to 24 hours per day, and never more than 10 Wh per clock hour. This avoids double-counting, but the typical-day panel only records hours, not what time of day each activity happened. So the tool would have to ask separately how many hours the connection is in use, or assume activities don't overlap.
- **(c) Per activity, with a note.** Count per activity as in (a), and state in the interface that overlapping activities overcount the router, by at most 10 Wh per overlapping hour.

Which do you want?

## 30. Learner (RG Sureshkumar)

---

go with option b

## Agent

---

Option (b) needs one more choice, because the panel records hours but not times of day.

**Question:** How should the tool know how many hours the router was actually in use?

- **(i) One extra field in the typical-day panel:** "Hours your home Wi-Fi is in use for these activities." It would start at the sum of the Wi-Fi activity hours, capped at 24, and the employee can lower it if activities overlap. For logged activities, overlap is calculated automatically from the log times.
- **(ii) No extra field.** The tool assumes typical-day activities don't overlap (sum of hours, capped at 24) and only removes overlap for logged activities, where times are known.

Option (i) lets people correct for overlap but adds a field. Option (ii) is simpler, but still overcounts typical-day overlap. Which do you want?

## 31. Learner (RG Sureshkumar)

---

go with option 1

## Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded the router rule: at most 10 Wh per clock hour, with the Wi-Fi hours field in the panel and overlap calculated from log times.

**Next: central values and ranges for devices.** The digital-day total needs a central value for each device, just as the AI total does. My proposal: - **Single values:** the phone (1 W), laptop (22 W), office desktop (115 W) and TV (100 W) have one figure each in S25 and no sourced range. They'd be labelled "single estimate, no range available", with no invented range added. - **Console gaming:** 214–219 W. The spread is so small that I'd use about 216 W as the central value, as the research already does. - **Gaming PC:** this one needs your decision. The PC-only figures come from two sources that disagree by about 50 W.

**Question:** What central value should the gaming PC use? The range is 319–388 W including the monitor at 14 W.

- **(a) About 373 W:** S36's 358.6 W, measured on one PC running one game, plus 14 W for the monitor. It's a real measurement but a single one, from a preprint.
- **(b) About 319 W:** S37's 305.1 W, a weighted average modelled from Steam hardware, plus 14 W. It covers many systems, but it's built from rated power, which S35 suggests overstates real draw.
- **(c) About 346 W:** the midpoint of the two. It's simple, but it's our own construction and would be labelled that way.

Which do you want?

## 32. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: Bash — Record device central values and read S31–S32]

```
S31. Greenly TikTok estimate, reported by Fortune (2024)

- Citation: Lake, S. (2024, December 13). Doomscrolling has reportedly made TikTok's annual carbon footprint almost the same as the entire country of Greece. *Fortune*.
<https://fortune.com/2024/12/13/tiktok-carbon-footprint-emissions-doomscrolling-greece-greenly>
- Claim or figure: one minute on TikTok emits 2.921 gCO2e; TikTok's annua ... [truncated]
```

I've recorded the central values: 373 W for the gaming PC, 216 W for console gaming, and single labelled figures for the other devices.

One correction to what I told you earlier. With the monitor's own 10–29 W range applied at both ends, the gaming PC range comes out at **315–388 W**, not the 319–388 W I quoted. The spec uses 315.

**Next: the data-centre share for social media.** Data centres are currently counted only for video streaming (about 1 Wh per hour, S25). The approved research accepted Meta's figure (S32) "for the data-centre share only": **about 5.5 kWh per daily active person per year, which is about 15 Wh per day**. It's a company-wide average across all of Meta's workloads, including AI and advertising, and it isn't tied to time spent.

**Question:** Should social media include a data-centre share from S32?

- **(a) Yes, about 15 Wh per day for anyone who uses social media**, whatever the hours. It would be labelled as Meta's company-wide per-person average, applied as a flat daily amount rather than per hour. This gives Robin and Alex a sourced server-side figure, but it doesn't scale with use, and it covers Meta only.
- **(b) No.** Social media counts the device and router only, and the interface says the server share is unknown. This is consistent with music, calls and gaming, which also have no data-centre figure, but it leaves out a sourced number that exists.

For scale, 15 Wh a day is about 15 hours of phone use at 1 W. Which do you want?

## 33. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: WebFetch — <https://www.cablelabs.com/blog/hourly-data-consumption-of-popular-video-conferencing-applications>]

```
Video Conferencing Data Consumption Findings
```

```
The article does not provide specific per-application figures for Google Meet, GoToMeeting, Microsoft Teams, and Zoom by name. Instead, the authors state: **"To avoid any appearance of endorsement of a particular conferencing application, we have not labeled the figures below with the specific apps under test."**
```

```
Available Data
```

```
Table 1 shows hou ... [truncated]
```

*[Tool: Bash — Read S22 Table 1 per-app values]*

```
Table 1 shows hourly bandwidth consumption (combining both upstream and downstream) for the laptop under test, normalized to Gigabytes per hour. The table provides the data consumption for the low-cost and higher-cost laptops in each scenario with the four conferencing applications. Table 1: Video Conferencing App Hourly Bandwidth Consumption in Gigabytes for Each User (Gigabytes/hour) The ... [truncated]
```

*[Tool: Edit — spec.md]*

```
The file spec.md has been updated successfully.
```

I've recorded the flat 15 Wh per day for social media, with its caveats.

**Next: a central value for video-call data.** I reread S22 (CableLabs). It tested four apps with 10-person gallery-view calls on two laptops, and found **0.5–3.4 GB per hour**. It deliberately doesn't say which app used how much ("App A–D"), and the per-app table is an image I can't read as text. So employees can't pick their app, and there's no sourced central value.

This matters most on cellular. At 0.1 kWh per GB, an hour's network share ranges from **50 Wh to 340 Wh**, far more than the laptop's 22 W. On home broadband it's 3–22 Wh.

**Question:** What central value should video-call data use?

- **(a) About 2 GB per hour**, the midpoint of the range, labelled as our own construction.
- **(b) The low end, 0.5 GB per hour**, with 3.4 GB as the high end of the range. This is the most conservative about the network share.
- **(c) The high end, 3.4 GB per hour**, with 0.5 as the low end.

Whichever you choose, the full 0.5–3.4 GB range is carried through as the low and high values. Which do you want?

## 34. Learner (RG Sureshkumar)

---

go with option a

**Agent**

---

[Tool: Bash — Compute digital-day worked example]

```
1.95 E 366.35 C g 139.2 W mL 1602.6
0.5 E 347.5 C g 132.1 W mL 1520.6
3.4 E 385.2 C g 146.4 W mL 1684.6
```

[Tool: Bash — Show current feature 3 draft]

```
Feature 3. Digital day builder
```

```
Draft for user review (not approved). Items marked **Open** still need the user's decision.
```

```
Need and outcome. The calculator cannot represent video calls, streaming, social media,
or gaming, although all three profiles spend substantial time on them, and every source
agrees that the device dominates their footprint (research §5–8). After this feature, an em
... [truncated]
```

[Tool: Bash — Complete feature 3 draft with checks]

```
1
```

Feature 3 is now fully drafted in [spec.md](#), under **Features** → **Feature 3**, with no open items. Video-call data uses a central value of 1.95 GB per hour, labelled as our own midpoint.

**Main worked check** (US region, nothing logged): - **Typical day**: 2 hours of calls on a laptop over Wi-Fi, 2 hours of HD streaming on a TV over Wi-Fi, and 1 hour of social media on a phone over cellular. - **Result**: about **366 Wh (348–385 Wh)**, **139 g CO<sub>2</sub>e** and **1.6 L of water**. The TV (200 Wh) dominates, followed by the network (about 64 Wh), the laptop (44 Wh) and the router (40 Wh).

These items are my proposals rather than your decisions, so please check them: - **Which devices each activity offers**. For example, gaming is offered only on a phone, console or gaming PC, and laptops aren't offered for gaming. - **Cellular is offered for the phone only**. - **The limitations list and the acceptance checks**. Check 7 says the digital-day total never affects the AI total, the budget or the team view, which follows from your Goal decisions.

**One loose end for planning**: when a logged activity replaces its typical-day entry, the home-Wi-Fi hours field would need to adjust as well. The spec states the intended behaviour, but not exactly how the field recalculates. I'd leave that detail to the planning stage unless you want it pinned down now.

**Question**: Does the Feature 3 draft look right, or is there anything to change before we move to feature 4, device manufacturing spread over use?

## 35. Learner (RG Sureshkumar)

---

looks good, lets move to feature 4

## Agent

---

## Feature 4: device manufacturing spread over use

The approved feature lets employees enter the devices they own and how long they keep them, and see manufacturing emissions spread over that time. Research left open whether manufacturing counts in totals at all.

**Evidence recap (research §9): - Manufacturing dominates.** Production is about 70–76% of life-cycle carbon for current phones, laptops and TVs (S39, S40, S42). - **Sourced figures cover:** a phone (iPhone 17, about 42 kg from production), a laptop (median 215 kg, P10–P90 158–287 kg, S41; MacBook Air about 110 kg from production), a TV (75-inch about 690 kg; a 55-inch proxy about 370 kg), and an office desktop (198 kg over its life cycle). - **Gaps:** there is no gaming-PC report, and consoles weren't researched. Manufacturing water isn't reported.

**Question:** How should manufacturing enter the totals?

- **(a) Added to the digital-day total** as a daily share: manufacturing carbon ÷ days the device is kept. The digital-day breakdown would show "manufacturing" as its own component.
- **(b) Shown beside the digital-day total but not added to it**, as a third figure. This keeps activity energy and manufacturing apart, since the device is owned whether or not it's used today, but it's one more number to explain.
- **(c) Not counted in any total.** Manufacturing is shown for information only. This is the simplest option, but it leaves out what the research says is most of a device's footprint, and it arguably wouldn't meet the brief's requirement that every feature contribute to the final calculation.

**For context:** the AI total already includes data-centre hardware manufacturing through EcoLogits' embodied carbon. Option (a) would treat the employee's own devices the same way. Which do you want?

## 36. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded that manufacturing is added to the digital-day total as its own component.

**Next: how manufacturing is spread over time.** Here's a worked example with the iPhone 17: about 42 kg from production, kept for 3 years.

- **(a) Per day kept:** 42 kg ÷ (3 × 365 days) is about **38 g CO<sub>2</sub>e per day**, however much the phone is used. The only input is how many years the device is kept. The research's illustrative spread over hours of use is a different calculation.
- **(b) Per hour of use:** the employee also enters typical daily hours for the device, say 3 hours, giving about **13 g per hour of use** (the research's illustrative figure). Each day's share then follows the hours

entered in the digital day (feature 3), so heavier use carries more manufacturing. That's closer to the approved wording, "spread over their use". But a device used 0 hours today would show no manufacturing, and it needs one more input per device.

(b) also raises a question: some devices, such as a laptop used for AI work, would be counted in the digital day only for the activities entered there.

**Question:** Which spreading method do you want?

## 37. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

*[Tool: Bash — Read device manufacturing sources S39–S44]*

```
S39. Apple, iPhone 17 Product Environmental Report (2025)

- Citation: Apple. (2025, September). *iPhone 17 Product Environmental Report*.
<https://www.apple.com/environment/pdf/products/iphone/iPhone_17_PER_Sept2025.pdf>
- Claim or figure: 55 kgCO2e (256GB) and 61 kgCO2e (512GB); breakdown for 256GB: production
materials and processes 53%, production electricity 23%, charging 18%, transportat ...
[truncated]
```

*[Tool: Edit — spec.md]*

*[Tool: Bash — Check Dell desktop report for manufacturing share]*

```
The file spec.md has been updated successfully.
```

```
dell.pdf: HTML document text, ASCII text
Shell cwd was reset to .
```

*[Tool: WebFetch — https://www.delltechnologies.com/asset/en-us/products/desktops-and-all-in-ones/technical-support/optiplex-tower-plus-7020-pcf-report.pdf]*

```
I cannot provide the requested breakdown because the document provided is a PDF file in raw
binary format, and the text content is not readable in the format submitted. The content
appears to be PDF metadata and encoded image/font data rather than accessible text.
```

```
To answer your question about the 198 kgCO2e lifecycle breakdown, I would need either:
- A properly extracted text version of the docu ... [truncated]
```

*[Tool: Bash — Extract text from Dell report PDF]*

```
Life Cycle Assessment
OptiPlex Tower Plus 7020
```

Report date: 2024 August

At Dell Technologies, we're working to reduce the impact of our business and products on the environment. Reducing the carbon footprint of our products is a critical consideration in our product designs and it also enables our customers to achieve their own sustainability goals.

This document summarizes the Life Cycle Asse ... [truncated]

I've recorded per-day spreading: manufacturing carbon ÷ (years kept × 365), with years kept entered for each device.

**Next: device manufacturing figures.** I reread S39–S44. I also opened Dell's report directly: its split by phase (manufacturing versus use) is only in a chart, so I can't read the desktop's manufacturing share as text.

Here's what the sources support:

| Device            | Proposed manufacturing carbon   | Default years kept | Source and caveat                                                                                                                       |
|-------------------|---------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Smartphone        | about 42 kg                     | 3                  | S39: 76% of the iPhone 17's 55 kg; one model                                                                                            |
| Laptop            | about 110 kg                    | 4                  | S40: 71% of the MacBook Air's 155 kg; one model. S41's 215 kg median (158–287) is cradle-to-grave including use, so it isn't comparable |
| TV, about 55-inch | about 370 kg                    | 7                  | S43: a commercial signage proxy, derived                                                                                                |
| TV, 75-inch       | about 690 kg                    | 7                  | S42                                                                                                                                     |
| Office desktop    | up to 198 kg (total life cycle) | 4                  | S44: the manufacturing share isn't published as text, so this overstates manufacturing, and it excludes the monitor                     |
| Gaming PC         | no figure                       | none               | no report found (research §9)                                                                                                           |
| Console           | no figure                       | none               | not researched                                                                                                                          |
| Monitor           | no figure                       | none               | not researched                                                                                                                          |

The default years come from each report's own assumption, and employees can change them.

**Question:** How should the three gaps (gaming PC, console, monitor) be handled?

- **(a) Search now** for manufacturer reports or LCAs, such as Sony PlayStation, monitor makers' product carbon footprints, or a gaming-desktop LCA. Anything verified goes into `research.md` for your assessment, and anything not found falls back to (b).
- **(b) Offer those devices with no manufacturing figure.** The interface says "no published figure found", so Jordan's gaming PC and monitor would add nothing.
- **(c) Use proxies.** For example, the office desktop's 198 kg as a stated lower bound for a gaming PC, labelled as a proxy.

Separately, please check whether the other rows look right to you.

## 38. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: WebSearch — search: PlayStation 5 product carbon footprint kgCO2e life cycle assessment Sony]

[Tool: WebSearch — search: Dell monitor product carbon footprint report 27 inch kgCO2e LCA P2725H]

[Tool: WebSearch — search: gaming desktop product carbon footprint report Alienware Aurora OR HP OMEN OR Lenovo Legion Tower kgCO2e PCF]

Web search results for query: "Dell monitor product carbon footprint report 27 inch kgCO2e LCA P2725H"

Links: [{"title": "Life Cycle Assessment P2725H Report date: March 2024", "url": "https://www.delltechnologies.com/asset/en-us/products/electronics-and-accessories/technical-support/p2725h-monitor-pcf-report.pdf"}, {"title": "s Dell S2721H Monitor From design to end-of-life and", "url": "https://dellte ... [truncated]"}]

Web search results for query: "PlayStation 5 product carbon footprint kgCO2e life cycle assessment Sony"

Links: [{"title": "PlayStation® and the environment | What we're doing to reduce our global impact", "url": "https://www.playstation.com/en-us/corporate/playstation-and-the-environment/"}, {"title": "Environmental impact of gaming products and services | University of Surrey", "url": "https://www.sur ... [truncated]"}]

Web search results for query: "gaming desktop product carbon footprint report Alienware Aurora OR HP OMEN OR Lenovo Legion Tower kgCO2e PCF"

Links: [{"title": "What Is The Carbon Footprint Of A Laptop? – Circular Computing", "url": "https://circularcomputing.com/news/carbon-footprint-laptop/"}, {"title": "GitHub – rarecoil/laptop-co2e: Carbon footprints for various developer-friendly laptops (notebook ... [truncated]"}]

[Tool: Bash — Fetch Dell monitor carbon reports and extract figures]

Exit code 1

m.pdf: PDF document, version 1.4, 3 pages

13:including the extraction of raw materials, production of components, transportation, use of the

18:Product Weight 8.47 kg Product Lifetime 5 yrs

20:Display Ratio 16:9 Manufacturing Location China

22:Ports DP,HDMI,VGA Use Location Europe

23:Total Carbon Footprint (EU Baseline): 194 kg CO2-eq. (excl. end of life credits)

24:Manufacturing brea ... [truncated]

[Tool: Bash — Extract Dell S2721H monitor datasheet text]

m2.pdf: PDF document, version 1.7, 2 pages  
s

Dell S2721H Monitor  
From design to end -of-life and everything in between, we work to improve the environmental impact of the products you purchase. As part of that process, we estimate the specific impacts throughout the lifecycle. This in ... [truncated]

[Tool: WebFetch — <https://www.playstation.com/en-us/corporate/playstation-and-the-environment/>]

[Tool: WebSearch — search: PlayStation 4 life cycle assessment 89 kg CO2e manufacturing study]

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

Web search results for query: "PlayStation 4 life cycle assessment 89 kg CO2e manufacturing study"

Links: [{"title": "Playing dangerously: The environmental impact of video gaming consoles", "url": "https://news.mongabay.com/2022/10/playing-dangerously-the-environmental-impact-of-video-gaming-consoles/"}, {"title": "What's the Carbon Footprint of Gaming Consoles? Xbox, Sony, Nintendo+", "url": "https:// ... [truncated]"}]

[Tool: WebFetch — <https://www.ifm.eng.cam.ac.uk/insights/sustainability/the-environmental-impact-of-a-playstation-4/>]

[Tool: WebFetch — <https://www.playstation.com/en-gb/corporate/playstation-and-the-environment/our-products/>]

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

```
PS4 Carbon Footprint Figure
```

```
Authors: Dr Claire Barlow and Dr John Durrell
```

```
Date: December 2019
```

```
Figure: 89 kilograms of CO2 per PS4
```

```
What it covers: Production and transportation of the console
```

```
Quote: "The equivalent of 89 kilograms of carbon dioxide is emitted into the atmosphere with the production and transportation of every PlayStation 4."
```

```
Method: Physical teardown ... [truncated]
```

I'm checking one more source that may cover console and gaming-PC manufacturing, then I'll report what I found.

*[Tool: WebFetch — <https://arxiv.org/abs/2608.19040>]*

```
Carbon Footprint Data for Gaming Hardware
```

```
Based on the webpage content provided, **I cannot locate specific embodied/manufacturing carbon figures (kgCO2e per unit) for gaming consoles or PCs.**
```

```
The page shows only the abstract, which states the paper "draws together publicly available data on game development, hardware, games sold, download sizes, time spent playing games on different platform ... [truncated]
```

*[Tool: Bash — Search Hot Games paper for embodied hardware figures]*

```
46716 hot.txt
```

```
Hot Games: Towards a Holistic Assessment of the Planet Warming Emissions of Video Games based on 2024–2025 Data // Restore the saved color scheme preference, or // enact the browser preference if "automatic", // without expecting DOM load to have completed. // // Also restore any saved readingmode and ToC display preferences. function initializeReadingPreferences() { let saved_theme ... [truncated]
```

*[Tool: Bash — Read Hot Games desktop, monitor, PS5 passages]*

```
for desktops, laptops, and dominant consoles. Steam Charts for PC playtime (Digital Ocean, 2026) . Sony reporting, scaled down for Xbox (Sony, 2025) . Switch neglected due to reports' lack of differentiation of handheld and docked mode (Nintendo Co., Ltd., 2026b) . PC 9.78 Sony "Game" 5.96 Xbox 1.99 Cloud gaming Industry reports for cloud gaming usage (Thakur, 2026) . Bitrates of cloud gam ... [truncated]
```

*[Tool: WebSearch — search: Microsoft Xbox Series X product carbon footprint report kg CO2e embodied 190]*

```
Web search results for query: "Microsoft Xbox Series X product carbon footprint report kg CO2e embodied 190"
```

```
Links: [{"title":"XBOX Sustainability Report | DitchCarbon","url":"https://ditchcarbon.com/organizations/xbox"}, {"title":"Gaming Sustainability – Microsoft Game Dev | Microsoft Learn","url":"https://learn.microsoft.com/en-us/xbox/sustainability/sustainability-overview"}, {"title":"How Micro ... [truncated]
```

*[Tool: WebSearch — search: "Xbox Series X" eco profile Microsoft PDF carbon footprint]*

Web search results for query: ""Xbox Series X" eco profile Microsoft PDF carbon footprint"

```
Links: [{"title":"How Microsoft and Xbox Plan to Transform Sustainable Gaming | Technology Magazine","url":"https://technologymagazine.com/articles/microsoft-making-xbox-maintenance-repair-more-sustainable"}, {"title":"An Update on Xbox Sustainability Efforts – XBOX Wire","url":"https://news.xbox.com/en-us/2 ... [truncated]
```

*[Tool: Bash — Extract Xbox Series X Eco Profile text]*

```
xbx.pdf: PDF document, version 1.7, 9 pages
E C O P R O F I L E
Xbox Series X
Xbox Series X Ecoprofile Last updated October 2023 © Microsoft Corporation. All rights reserved.
Our Commitments
Microsoft is committed to becoming carbon negative, water positive, and zero waste by 2030. Our mission as Team Xbox is to bring the joy and community of gaming to everyone on the planet, but it's critica ... [truncated]
```

I've finished the search for the three gaps. Here's what I found and verified:

**Console: one verified figure.** Microsoft's [Xbox Series X Eco Profile](#) (October 2023) gives **190 kg CO<sub>2</sub>e for production**, out of 729 kg over its full life cycle with 5 years of use. I read this directly. - Sony publishes nothing for the PS5. - A 2019 Cambridge teardown of a PS4 ([IfM](#)) estimated 89 kg for production and transport. It's older hardware and a blog-style summary, not a paper.

**Gaming PC: no report exists.** A recent workshop paper ([Hazas et al., "Hot Games", arXiv:2608.19040](#)) confirms this. It used an office desktop as a stand-in: Dell Vostro 3030, 112 kg for manufacturing and transport. It calls that "conservative" because high-end systems "can exceed 200 kg".

**Monitor: sources disagree sharply.** Both figures below are for Dell 27-inch monitors: - **S2721H (2021, Dell's older PAIA tool):** 523 ± 100 kg over the life cycle, 63.9% manufacturing, so about **334 kg** from manufacturing. - **P2725H (2024, Dell's newer Sphera-based tool): 194 kg for the whole life cycle.** The manufacturing share is only in a chart.

The gap seems to come from Dell changing its method, not from the monitors themselves.

**My proposal:** - **Console:** 190 kg (Xbox Series X), labelled as a stand-in for all consoles because Sony publishes nothing. Default 5 years. - **Gaming PC:** 112 kg ("Hot Games" office-desktop proxy), labelled as a

conservative lower bound. Default 4 years. - **Monitor:** 194 kg, the newer Dell figure, labelled as a whole life cycle total because the manufacturing share can't be separated. The older 334 kg would be disclosed alongside it. Default 5 years.

Each would be added to `research.md` for your assessment.

**Question:** Do you want to use these three proposals, or change any of them? The monitor is the least certain: 194 kg is newer but includes use-phase emissions, while 334 kg is manufacturing only but uses Dell's older method.

## 39. Learner (RG Sureshkumar)

---

use those three proposals

### Agent

---

[Tool: Bash — Get Hot Games author list]

```
citation_author" content="Hazas, Mike"
citation_author" content="Dalli, Kevin C."
citation_author" content="Menon, Arjun"
citation_author" content="Abraham, Ben"
citation_author" content="Nordgren, Ossian"
Shell cwd was reset to .
```

[Tool: Edit — `research.md`]

[Tool: Edit — `spec.md`]

The file `research.md` has been updated successfully.

The file `spec.md` has been updated successfully.

I've added the five new sources to `research.md` (S62–S66, all marked not yet assessed) and put the full device table in the `spec`.

**Next: energy and water for manufacturing.** Every manufacturing source we're using reports **carbon only**: - **Energy:** none of the reports gives manufacturing energy in Wh. - **Water:** research §9 found none of these reports quantifies manufacturing water. The only phone water figure (S51, about 12,100 L) is mostly grey water, a notional dilution volume. Using it would break the blue-water consistency that feature 5 is meant to fix.

**Question:** How should manufacturing appear when the page shows energy or water instead of carbon?

- **(a) Carbon only.** Manufacturing is added to the digital-day total in carbon. In the energy and water views it is left out, with a visible note saying no comparable figure exists. This is honest, but the digital-day total differs in scope between metrics.

- **(b) Carbon, plus S51's water for the phone only**, labelled as mostly grey water. Other devices have no water figure. This is inconsistent across devices and with feature 5's blue-water approach.

Which do you want?

## 40. Learner (RG Sureshkumar)

---

go with option b

### Agent

---

[Tool: Bash — Read S51 smartphone water source]

```
S51. GRACE Communications, Water Footprint Calculator: "The Hidden Water in Everyday Products"

- Citation: GRACE Communications Foundation. The Hidden Water in Everyday Products. *Water Footprint Calculator*. <https://watercalculator.org/footprint/the-hidden-water-in-everyday-products/>
- Claim or figure: a smartphone's production water footprint is an estimated 3,190 gallons, with the grey w ... [truncated]
```

[Tool: Edit — spec.md]

```
The file spec.md has been updated successfully.
```

I've recorded that: carbon for every device, the S51 phone water figure labelled as mostly grey water, and no manufacturing energy.

**Scale to be aware of:** the phone's water share alone is about **11 L per day**. That's roughly seven times the whole example digital day in feature 3 (about 1.6 L). So in the water view it will dominate, and it's measured differently from everything else. The spec says the interface must state that plainly. It'll come up again in feature 5.

**Next: how employees enter their devices.**

**Question:** How should the device list be built?

- **(a) A separate "My devices" list.** Each row has a device type, the number owned, and years kept (pre-filled with the default). It's independent of the digital-day activities, so it includes devices the employee owns but didn't use today, which fits the per-day-kept method.
- **(b) Built automatically from the digital-day activities.** One of each device used in feature 3 appears, with years kept editable. There's less typing, but devices not used in an activity row, such as a monitor or a second phone, would be missed unless added by hand.

A related question for after this one: should a shared device, such as a household TV, be split between the people who use it?

## 41. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded the "My devices" list: device type, number owned and years kept, with the defaults pre-filled.

**Shared devices.** Take a household TV (about 370 kg to manufacture, kept 7 years). It adds about **145 g CO<sub>2</sub>e per day** if one person counts all of it. That's more than the whole example digital day's electricity carbon (about 139 g). If three people share it, each person's share is about 48 g.

**Question:** Should shared devices be split?

- **(a) Yes, with an optional "shared with how many people" field on each device row**, defaulting to 1. Each person counts their share. It's more accurate for households, but it's one more field, and it relies on self-reporting.
- **(b) No.** Each employee counts devices they own in full. This is simpler, but it overstates manufacturing for anyone who shares a TV or other household device, and the interface would say so.

Which do you want?

## 42. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: Bash — Compute feature 4 checks and find existing device citations]

```
38.4 75.3 48.3 162.0 11.03 144.8
347: <a href="https://www.epa.gov/greenvehicles/greenhouse-gas-emissions-typical-
passenger-vehicle" target="_blank" rel="noopener">EPA's ~400 g CO₂/mile
385: a <a href="https://www.epa.gov/sites/default/files/2018-
02/documents/lca_tv.pdf" target="_blank" rel="noopener">new flat-screen TV ≈ 350 kg;
386: a <a href="https://deve ... [truncated]
```

[Tool: Bash — Complete feature 4 draft with checks]

1

Feature 4 is now fully drafted in [spec.md](#), under **Features** → **Feature 4**, with no open items.

**How it's calculated:** each device's daily share is its manufacturing carbon × number owned ÷ (years kept × 365) ÷ people sharing. The shares are added into the digital-day total as a "manufacturing" component.

**Main worked check:** a phone (3 years), a laptop (4 years) and a 55-inch TV (7 years, shared by 3) come to **about 162 g CO<sub>2</sub>e per day**: phone 38.4 g, laptop 75.3 g, TV 48.3 g. In the water view, the phone adds about 11 L per day, labelled as mostly grey water.

Please check one proposal of mine in particular. The calculator's current yearly comparison items quote outdated or unsupported figures: a new smartphone at 70 kg, a laptop at 250 kg, and a TV at 350 kg, whose EPA citation doesn't contain that number. I've written that feature 4 replaces them with the new figures (**42 kg, 110 kg and 370 kg**) and their sources. The research selected feature 4 partly for this reason. Their water figures are left to feature 5.

**Question:** Does the Feature 4 draft look right, or is there anything to change before we move to feature 5, like-for-like comparisons?

### 43. Learner (RG Sureshkumar)

---

feature 4 is good, lets move to feature 5

### Agent

---

*[Tool: Bash — Inspect comparison data and water methodology text]*

```
// Blue-water comparisons (gal): freshwater actually withdrawn, green rain excluded.
// Irrigated crops (almonds, cotton) and lawns dominate; rain-fed foods nearly vanish.
AIPF.DAILY_WATER_ITEMS = [
 { label: 'A cup of coffee', w: 1 },
 { label: 'A beef burger', w: 1.6 },
 { label: 'A slice of bread', w: 2 },
 { label: 'An egg', ... [truncated]
```

*[Tool: Bash — Check where water comparison values are used]*

```
assets/js/core.js:148: return ((loc.w + home.w + drive.w + diet.w + fly.w) * GAL_TO_L) /
DAYS;
assets/js/core.js:150: function itemBase(item, metric) { return metric === 'carbon' ?
item.c * 1000 : item.w * GAL_TO_L; }
assets/js/core.js:158: function linesForSize(s) { return Math.round((s.w /
TOKENS_PER_WORD) * CODE_FRACTION / TOKENS_PER_LINE); }
assets/js/calculator.js:252: const strip = (... [truncated]
```

*[Tool: Bash — Trace where full-footprint water values are used]*

```
assets/js/core.js:150: function itemBase(item, metric) { return metric === 'carbon' ?
item.c * 1000 : item.w * GAL_TO_L; }
assets/js/core.js:171: aiDailyEnergy, dailyFootprint, itemBase,
isCodeRow, linesForSize,
```

```
assets/js/calculator.js:218: const v = AIPF.itemBase(it, state.metric);
 }
 AIPF.renderBars = renderBars;

 function renderDaily() {
 const ai = AIPF.aiD ... [truncated]
```

Before we decide anything on feature 5, I checked the current code. The problem is narrower than `research.md` describes, and part of it is in a different place.

**What's already right.** The water charts don't use the burger's 1,700 L or the coffee's 140 L. The code already has separate **blue-water** comparison lists (`DAILY_WATER_ITEMS` and `ANNUAL_WATER_ITEMS` in [data.js](#)), and those are what the water charts draw. The full-footprint figures appear only in the methodology text of [index.html](#).

**What's still inconsistent:**

1. **The burger.** The chart uses about **6 L** of blue water per burger, with no source given. S50's figures (beef 15,000 L/kg, 4% blue) suggest about **70 L**, as research §10 derived. That's roughly a 10× disagreement.
2. **The smartphone.** The yearly water chart uses **3,370 gal (about 12,760 L)**, described as "mostly ultrapure freshwater", but it cites a phone-recycling blog. S51 describes a similar total as **mostly grey water**. It's in a blue-water chart without being blue water.
3. **Electricity.** "A day of home electricity" and "an hour of air conditioning" use NREL's **0.47 gal/kWh (about 1.8 L/kWh)**. The AI total and the digital-day total use S46's **4.35–4.88 L/kWh**. So the same electricity is counted with different water factors on the same page.
4. **Other blue values:** coffee about 4 L, egg, rice, almonds and so on. Their sources haven't been checked against the figures used.
5. **The methodology text** still quotes the full-footprint figures (1,700 L for a burger, 140 L for coffee, 12,800 L for a phone) next to the blue-water charts.
6. **Carbon comparisons haven't been examined yet.** For example, whether they use the same kind of boundary as the AI figures.

Also, the comparisons currently sit next to the AI total only. There's now a digital-day total too.

**Question:** What should feature 5 cover?

- **(a) Water only.** Verify and source every blue-water comparison item, correct or remove those that aren't blue (the smartphone), bring electricity items to the S46 factor, and fix the methodology text.
- **(b) Water and carbon.** Everything in (a), plus a check of each carbon comparison's boundary and source against the AI and digital-day figures.
- **(c) Either (a) or (b), plus visible labels** in the charts showing each comparison item's boundary (for example "blue water, consumption" or "life cycle").

Separately, should the comparisons also be shown next to the digital-day total, or stay with the AI total only?

## 44. Learner (RG Sureshkumar)

---

go with option b

## Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded the scope (water and carbon) and the six issues I found, as a "current state" note in the Feature 5 draft.

Checking every item means about 30 comparison items in total. We need a rule before I start, so that each item is judged the same way.

**Proposed rule:** - **Water items** must use a sourced **blue-water consumption** figure, the same kind of water as the AI and digital-day figures. Items built on electricity use S46's 4.35 L/kWh. For example, a day of home electricity (about 29 kWh) would become about 126 L instead of about 53 L. - **Carbon items** must use a sourced figure whose boundary is stated. That's either life cycle (for example a burger or jeans) or electricity only. Electricity-based items (the microwave, dryer and so on) use kWh × the selected region's grid, as the AI figures do. - **Items that can't be verified to that standard are removed**, not kept with a caveat. The methodology text is rewritten to match the figures actually used.

**Question:** Do you want to use this rule? Or would you rather keep unverifiable items with a visible "unverified" label instead of removing them?

## 45. Learner (RG Sureshkumar)

---

use this rule

## Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded the rule.

The question from before is still open: the everyday comparisons (daily bars and yearly "ways you add or cut" charts) currently sit next to the AI total only. Now there's also a digital-day total.

**Question:** Where should the comparisons appear?

- **(a) Beside both totals.** The AI total and the digital-day total each appear as a highlighted bar in the comparison charts, next to each other, not combined, consistent with the Goal. Employees can then see both against everyday items, which is the brief's aim of placing AI "alongside digital habits and other everyday activities".

- **(b) AI total only, as now.** The digital-day total is shown only as its own figure. This is simpler, but digital habits never get the everyday context.

Which do you want?

## 46. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've recorded that: both totals appear as separate highlighted bars in the comparison charts.

**Last open item for feature 5: when to check the roughly 30 comparison items.**

**Question:** When should I do the item-by-item verification?

- **(a) Now, in this specification stage.** I'd check each item's source against the rule and record the results in research.md for your assessment. The spec would then list exactly which items are kept (with figures), corrected or removed, so the spec is fully concrete. This is a long stretch of source-checking and will make this chat considerably longer.
- **(b) At the start of implementation.** The spec fixes the rule and acceptance checks, such as "every comparison item has a verified source with a stated boundary". The first implementation step then does the checking and brings the results to you for approval before any code changes. This conversation stays shorter, but the spec wouldn't yet say which items survive.

Which do you want?

## 47. Learner (RG Sureshkumar)

---

go with option a

### Agent

---

[Tool: Bash — Read water sources S49–S50 and count page links]

### S49. Dieter & Maupin, USGS (2017)

– Citation: Dieter, C. A., & Maupin, M. A. (2017). \*Public Supply and Domestic Water Use in the United States, 2015\* (Open-File Report 2017–1131). U.S. Geological Survey.

<<https://pubs.usgs.gov/of/2017/1131/ofr20171131.pdf>>

– Claim or figure: national average total domestic per capita use was 82 gallons per capita per day in 2015 (down from 88 in 2010), rangi ... [truncated]

Next I'm checking the primary water-footprint datasets (Mekonnen & Hoekstra), which most of the food blue-water figures should trace back to.

*[Tool: Bash — Download Mekonnen & Hoekstra 2011 crops paper and search products]*

```
mh2011.pdf: PDF document, version 1.4
42:with relatively large water footprints are: coffee, tea, cocoa,
55:ter footprint was calculated for wheat (1087 Gm3 yr-1), rice
56:(992 Gm3 yr-1) and maize (770 Gm3 yr-1). Wheat and rice
395:footprint are rice (992 Gm3 yr-1) and maize (770 Gm3 yr-1).
405:footprint is lowest for date palm (43 %) and cotton (64 %).
407:duction was 899 Gm3 yr-1. Wheat (204 Gm ... [truncated]
```

*[Tool: Bash — Read Table 3 headings and coffee, avocado rows]*

```
and California (20 Gm3 yr-1) in the USA. Large grey water
footprints were estimated for China, the USA and India.
Hydrol. Earth Syst. Sci., 15, 1577–1600, 2011 www.hydrol-earth-syst-sci.net/15/1577/2011/
M. M. Mekonnen and A. Y. Hoekstra: The green, blue and grey water footprint of crops 1585
Table 3. Global average water footprint of primary crops and derived crop products. Period:
1996–2005.
FA ... [truncated]
```

*[Tool: Bash — Find milled rice and fetch animal products report]*

```
https://www.waterfootprint.org/resources/Report48-WaterFootprint-AnimalProducts-Vol1.pdf
mh2012.pdf: HTML document text, ASCII text, with very long lines (4894), with CRLF, LF line
terminators
https://waterfootprint.org/media/downloads/Mekonnen-Hoekstra-2012-
WaterFootprintFarmAnimalProducts.pdf
mh2012.pdf: PDF document, version 1.6 (zip deflate encoded)
https://www.waterfootprint.org/resources/Mek ... [truncated]
```

*[Tool: Bash — Extract animal products paper and find beef, egg, milk]*

```
A Global Assessment of the Water
Footprint of Farm Animal Products
Mesfin M. Mekonnen* and Arjen Y. Hoekstra
Department of Water Engineering and Management, University of Twente, P.O. Box 217, 7500 AE
Enschede, The Netherlands
ABSTRACT
17:The average water footprint per calorie for beef is
19:The water footprint per gram of protein for milk,
20:eggs and chicken meat is 1.5 times larger than for
173 ... [truncated]
```

*[Tool: Bash — Read animal products water footprint table]*

```
tries. Due to the specific climate and poor agricul-
tural practices in India, the water footprint per ton
```

of feed in this country is larger than in the Netherlands and the USA.  
When we look at global averages (Table 1), we see that the water footprint of meat increases from chicken meat (4,300 m<sup>3</sup>/ton), goat meat (5,500 m<sup>3</sup>/ton), pig meat (6,000 m<sup>3</sup>/ton) and sheep meat (10,400 m<sup>3</sup>/ton) to beef (1 ... [truncated]

*[Tool: Bash — Download paper water footprint reports]*

Report46-WaterFootprintPaper.pdf: PDF document, version 1.6, 23 pages (zip deflate encoded)  
Report64-WaterFootprint-Paper.pdf: HTML document text, ASCII text, with very long lines (4894), with CRLF, LF line terminators  
Shell cwd was reset to .

I'm checking the water-footprint sources behind the comparison items. So far I've verified the food figures against the primary datasets. Now I'm on the paper (book-printing) figure.

*[Tool: Bash — Search paper report for blue water per sheet]*

Value of Water Research Report Series No .46  
The green and blue water  
footprint of paper products:  
Methodological considerations  
and quantification  
Value of Water  
P.R. van Oel  
A.Y. Hoekstra  
July 2010

THE GREEN AND BLUE WATER FOOTPRINT OF PAPER PRODUCTS:  
METHODOLOGICAL CONSIDERATIONS AND QUANTIFICATION

P.R. VAN OEL1  
16:THE GREEN AND BLUE WATER FOO ... [truncated]

*[Tool: Bash — Read paper industrial-stage blue water passage]*

Norway 1051 971 1181  
Portugal 1658 2017 848  
Spain 598 728 301  
South Africa 755 702  
Austria 826 1004  
New Zealand 867 909 874  
Australia 993 823 623 657

```
Poland 1230 1048
Thailand 759
```

Water footprint of paper products in industrial stage – example USA  
In the USA, annual industrial production of paper products is around 97×106 ton/year. The total water use for the  
m ... [truncated]

[Tool: WebFetch — <https://www.eia.gov/tools/faqs/faq.php?id=97>]

```
US Residential Electricity Consumption

According to the EIA, "In 2022, the average annual amount of electricity sold to (purchased by) a U.S. residential electric-utility customer was 10,791 kilowatthours (kWh), an average of about 899 kWh per month."
```

[Tool: WebFetch — <https://www.siliconvalleypower.com/residents/save-energy/appliance-energy-use-chart>]

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

[Tool: Bash — Fetch appliance chart with browser user agent]

```
ugrep: error: error at position 135
:[^\n\x80-\xbf][\x80-\xbf]*){0,120}
 exceeds complexity limits___/
```

[Tool: Bash — Search appliance chart text for items]

```
412 svp.txt
Shell cwd was reset to .
```

I've finished checking the **daily water** comparisons, the first of four groups. The results need two decisions from you before I go further.

**Sources I verified directly:** - Mekonnen & Hoekstra (2011), HESS 15:1577, Table 3: global average blue water for crop products. - Mekonnen & Hoekstra (2012), *Ecosystems* 15:401, Table 1: blue water for animal products. - **EIA's household electricity FAQ:** 10,791 kWh per year in 2022, about 29.6 kWh per day.

| Item                        | Current     | Verified blue water            | Status     |
|-----------------------------|-------------|--------------------------------|------------|
| Cup of coffee (7 g roasted) | about 3.8 L | <b>about 1.0 L</b> (139 L/kg)  | correct it |
| Beef burger (113 g)         | about 6 L   | <b>about 62 L</b> (550 L/kg)   | correct it |
| Slice of bread (30 g)       | about 7.6 L | <b>about 9.0 L</b> (301 L/kg)  | correct it |
| Egg (60 g)                  | about 15 L  | <b>about 14.6 L</b> (244 L/kg) | matches    |

| Item                                   | Current     | Verified blue water                    | Status                     |
|----------------------------------------|-------------|----------------------------------------|----------------------------|
| Glass of milk (250 g)                  | about 23 L  | <b>about 21.5 L</b> (86 L/kg)          | matches                    |
| Bowl of rice (75 g dry, husked)        | about 26 L  | <b>about 33 L</b> (443 L/kg)           | correct it                 |
| Avocado (200 g)                        | about 61 L  | <b>about 57 L</b> (283 L/kg)           | matches                    |
| Handful of almonds (28 g)              | about 121 L | <b>about 107 L</b> (3,816 L/kg)        | correct it                 |
| A day of home electricity (29.6 kWh)   | about 53 L  | <b>about 129 L</b> (× 4.35 L/kWh, S46) | correct it                 |
| Hour of air conditioning (about 3 kWh) | about 5.3 L | about 13 L if 3 kWh is sourced         | <b>no source</b> for 3 kWh |
| Printing a book                        | about 19 L  | not verifiable (see below)             | remove it                  |

**The book:** the paper study (van Oel & Hoekstra 2010) doesn't separate green from blue water for forestry, only a partial industrial figure. Under your rule, it's removed.

**The air conditioner:** the page gives no source for "about 3 kWh". Separately, the microwave's cited source (Silicon Valley Power) now blocks automated access, so I can't check it. I'll deal with both in the carbon group.

**First decision: global or US averages.** The table uses **global** averages, because food is traded worldwide and the AI water can't be tied to a location either. US-specific values exist for animal products: beef 525 L/kg, egg 130 L/kg, milk 60 L/kg. The egg would drop to about 7.8 L.

**Second decision: serving sizes.** The masses in brackets (113 g burger, 30 g slice and so on) are **our own stated assumptions**, not sourced facts. They'd be shown in the interface as "per 113 g patty" and so on.

**Question:** Do you want global averages, with serving sizes shown as stated assumptions? Or would you rather make a different choice on either point?

## 48. Learner (RG Sureshkumar)

use global averages

## Agent

[Tool: Bash — Try to download cotton water footprint report]

```
https://waterfootprint.org/resources/Report18.pdf: PDF document, version 1.6, 17 pages
https://www.waterfootprint.org/resources/Report18.pdf: PDF document, version 1.6, 17 pages
https://waterfootprint.org/media/downloads/Report18.pdf: PDF document, version 1.6, 17
pages
https://waterfootprint.org/media/downloads/Chapagain_et_al_2006_cotton.pdf: HTML document
text, ASCII text, with ver
Shell cwd wa ... [truncated]
```

[Tool: Bash — Search cotton report for T-shirt, jeans, blue share]

```
head: r18.txt: No such file or directory
ugrep: warning: r18.txt: No such file or directory
Shell cwd was reset to .
```

*[Tool: Bash — Extract cotton report text and search]*

```
invalid pdf header: b'<!DOC'
EOF marker not found
Traceback (most recent call last):
 File "<string>", line 3, in <module>
 File "~/Library/Python/3.9/lib/python/site-packages/pypdf/_reader.py", line 151, in
__init__
 self._initialize_stream(stream)
 File "~/Library/Python/3.9/lib/python/site-packages/pypdf/_reader.py", line 175, in
_initialize_stream
 self.read(stream)
 File "~/Library/ ... [truncated]
```

*[Tool: Bash — Re-download and search cotton report]*

```
The water footprint of
cotton consumption
Value of Water
A.K. Chapagain
A.Y. Hoekstra
H.H.G. Savenije
R. Gautam
September 2005
39:4. The value of irrigation water in Nyanya dzi smallholder irrigation scheme, Zimbabwe.
76:2. Green, blue and dilution water
..... 11
96:between three types of impact: evaporation of in ... [truncated]
```

*[Tool: Bash — Read cotton report summary and product table]*

#### Summary

The consumption of a cotton product is connected to a chain of impacts on the water resources in the countries where cotton is grown and processed. The aim of this report is to assess the 'water footprint' of worldwide cotton consumption, identifying both the location and the character of the impacts. The study distinguishes between three types of impact: evaporation of infiltration ... [truncated]

I'm on the yearly water items now. The cotton figures are verified, and next I'm checking the EPA lawn-watering figure.

*[Tool: WebFetch — <https://www.epa.gov/watersense/landscaping-tips>]*

Based on my review of the provided EPA WaterSense landscaping tips page, I cannot find any specific figures about:

- Water usage for average households or lawns (in gallons per year or day)
- Water loss percentages from overwatering or evaporation

- Quantified savings from native plants or reduced watering practices

The page provides qualitative guidance (such as "native plants require little wa ... [truncated]

[Tool: WebFetch — <https://www.epa.gov/watersense/outdoors>]

#### # Outdoor Water Use Statistics

Based on the EPA WaterSense page content provided, here is the only specific figure given:

**\*\*"residential outdoor water use across the United States accounts for nearly 8 billion gallons of water each day, mainly for landscape irrigation"**

Additionally, the page states: **\*\*"The average U.S. household uses more water outdoors than for showering and washing clothes ... [truncated]**

I've finished the **yearly water** group, the second of four.

**New source verified directly:** Chapagain et al. (2005), *The water footprint of cotton consumption* (Water Footprint Network, Value of Water Report 18), Table 3.5. It gives global average blue water for **final cotton textile: 4,917 L/kg**. Its blue water is irrigation and processing water withdrawn.

| Item                                             | Current              | Verified or derived blue water                                                   | Status                               |
|--------------------------------------------------|----------------------|----------------------------------------------------------------------------------|--------------------------------------|
| A year of daily coffee                           | about 1,380 L        | <b>about 355 L</b> (365 × 0.97 L)                                                | correct it                           |
| A cotton T-shirt (250 g)                         | about 1,500 L        | <b>about 1,230 L</b>                                                             | correct it                           |
| A pair of jeans (800 g)                          | about 5,000 L        | <b>about 3,930 L</b> (cotton only, excludes processing beyond the textile stage) | correct it                           |
| A year of daily almonds                          | about 44,300 L       | <b>about 39,000 L</b> (365 × 107 L)                                              | correct it                           |
| Five fewer cotton garments (1 year, as T-shirts) | about 7,570 L saved  | <b>about 6,150 L</b>                                                             | correct it                           |
| <b>A new smartphone</b>                          | about 12,760 L       | mostly grey water (S51); its source is a recycling blog                          | <b>remove it</b> from the blue chart |
| <b>Replacing a lawn with native plants</b>       | about 56,800 L saved | EPA's pages give no per-lawn figure                                              | <b>remove it</b>                     |
| <b>Letting your lawn go unwatered</b>            | about 75,700 L saved | as above                                                                         | <b>remove it</b>                     |

On the lawns: [EPA WaterSense](#) says only that US residential outdoor use is "nearly 8 billion gallons of water each day". Turning that into a per-household figure would mean dividing by a household count from another source, and it would measure water withdrawn, not consumed. I'd treat that as not meeting your rule.

**The effect:** the yearly **"ways to cut water"** chart would be left with a single item, five fewer cotton garments. That's thin, but it's honest.

Garment masses (250 g T-shirt, 800 g jeans) are our stated assumptions, like the serving sizes.

**Question:** Do you accept these corrections and removals? Or would you rather I try to build a sourced lawn figure (the EPA total ÷ a Census household count, labelled as derived and as withdrawal), so the "cut" chart has more than one item?

## 49. Learner (RG Sureshkumar)

---

accept the corrections and removals

### Agent

---

*[Tool: Edit — spec.md]*

*[Tool: Edit — research.md]*

The file spec.md has been updated successfully.

The file research.md has been updated successfully.

*[Tool: Bash — Read carbon comparison methodology text]*

```
the rarely / sometimes / often options are roughly 0.5, 1.5, and 5 such trips a year.
 The water footprint uses blue water only –
freshwater
 actually drawn from rivers, lakes, and aquifers. We deliberately leave out
 green water (rain that falls on farmland and would evaporate anyway,
whether or
 not a crop is grown) and g ... [truncated]
```

*[Tool: WebFetch — <https://theconversation.com/heres-how-much-water-it-takes-to-make-a-serving-of-beef-and-why-where-it-comes-from-is-so-important-208155>]*

```
Water Footprint of Beef: Key Figures

Per Kilogram of Beef:
- "more than 15,000 litres" total water footprint
- 67 litres of blue water (UK national average)
- Nearly 2,000 litres of blue water (US irrigated systems)

Per Serving (375g English topside):
- "33 litres of blue water, 96% of which goes towards feeding and raising the animal"

Source: Tim Hess and Adrian Williams, Cranfie ... [truncated]
```

*[Tool: WebFetch — <https://www.epa.gov/greenvehicles/greenhouse-gas-emissions-typical-passenger-vehicle>]*

*[Tool: WebFetch — <https://iopscience.iop.org/article/10.1088/1748-9326/aa7541>]*

*[Tool: WebFetch — <https://ourworldindata.org/grapher/ghg-per-kg-poore>]*

```
EPA Greenhouse Gas Emissions Data
```

```
Grams of CO2 per mile: "The average passenger vehicle emits about 400 grams of CO2 per mile."
```

```
Annual miles: "drives around 11,500 miles per year"
```

```
Annual tonnes CO2: "A typical passenger vehicle emits about 4.6 metric tons of CO2 per year."
```

```
Tailpipe CO2 only: Yes, these figures represent tailpipe emissions only. The page notes separately that ... [truncated]
```

I appreciate your request, but I cannot provide the specific greenhouse gas emissions values for Beef (beef herd), Beef (dairy herd), and Coffee from this webpage because **the actual numerical data is not included in the content provided**.

The page only contains:

- A description explaining that emissions are measured in "kgCO<sub>2</sub>eq per kg"
- Information about the data source: "Poore and Nemecek (2 ... [truncated])"

```
Emission Reductions from Wynes & Nicholas (2017)
```

Based on Figure 1 and Table 1 in the paper published in *Environmental Research Letters*, Volume 12, Number 7 (July 2017), DOI 10.1088/1748-9326/aa7541:

```
High-Impact Actions:
```

- "living car free" (car-free lifestyle): **2.4 tCO<sub>2</sub>e saved per year** (from abstract)
- Avoiding one transatlantic round-trip flight: **1.6 tCO<sub>2</sub>e saved per roundtrip** ... [truncated]

*[Tool: Bash — Download Poore & Nemecek data via Our World in Data]*

```
Entity,Year,Greenhouse gas emissions per kilogram
Apples,2010,0.43
Beef (beef herd),2010,99.48
Beef (dairy herd),2010,33.3
Coffee,2010,28.53
Eggs,2010,4.67
Milk,2010,3.15
Soy milk,2010,0.98
Shell cwd was reset to .
```

I've confirmed the car and diet figures. Next I'm checking the appliance, book, clothing and household-goods sources in one pass.

*[Tool: WebFetch — [https://www.energystar.gov/products/clothes\\_dryers](https://www.energystar.gov/products/clothes_dryers)]*

*[Tool: WebFetch — <https://www.energystar.gov/products/dishwashers>]*

*[Tool: WebFetch — <https://howbadarebananas.com/>]*

[Tool: WebFetch — <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1530-9290.2011.00414.x>]

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

#### # About This Site

This is a website promoting the book *\*How Bad Are Bananas?\**, which examines the environmental impact of everyday activities and purchases.

**\*\*Does it provide specific carbon footprints?\*\***

No specific figures are given in this page content. The site describes the book's general approach: "what's the real impact of each of the things we do and buy?" and mentions it compares items ... [truncated]

#### # Electricity Use Figures for Clothes Dryers

I reviewed the provided ENERGY STAR webpage about clothes dryers, but **\*\*it does not contain specific electricity consumption figures\*\*** (in kWh per load or per year) comparing conventional dryers to ENERGY STAR models.

The page does reference one financial metric:

"A full-size, electric clothes dryer that has earned the ENERGY STAR saves approximately ... [truncated]

#### # ENERGY STAR Dishwasher Efficiency Figures

Based on the webpage content provided, I cannot locate specific electricity consumption figures (kWh per year or per cycle) or water usage per cycle for either ENERGY STAR or standard dishwashers.

The page does mention these related figures:

**\*\*Annual water savings:\*\*** "A standard-size ENERGY STAR certified dishwasher...can save you about 5,800 gallons ... [truncated]

[Tool: Bash — Check book LCA metadata and download Levi's LCA deck]

```
['Carbon Footprint Assessment of a Paperback Book'] [{'given': 'Jean-Robert', 'family': 'Wells', 'sequence': 'first', 'affiliation': [], 'role': [{'vocabulary': 'crossref', 'role': 'author'}]}, {'given': 'Jean-François', 'family': 'Boucher', 'sequence': 'additional', 'affiliation': [], 'role': [{'vocabulary': 'crossref', 'role': 'author'}]}, {'given': 'Achille-Benjamin', 'family': 'Laurent', 'sequ ... [truncated]
```

[Tool: Bash — Search Levi's LCA deck for jeans carbon]

15: billion liters of water saved to date through the Levi's® Water<Less™ process and implementation of the apparel  
122: CLIMATE CHANGE Global warming potential of greenhouse gases released to the environment kg CO<sub>2</sub>-e  
123: WATER INTAKE Freshwater taken from the environment liters  
124: WATER CONSUMPTION Net freshwater taken from the environment minus water returned  
126: liters  
147: • Water C ... [truncated]

[Tool: WebFetch — <https://iopscience.iop.org/article/10.1088/1748-9326/ab8589>]

[Tool: WebFetch — <https://wwwFOUNDERSpledge.com/research/climate-and-lifestyle-report>]

#### # Climate & Lifestyle Report – CO<sub>2</sub> Emissions Figures

\*\*Report Date:\*\* February 10, 2020  
\*\*Author:\*\* Johannes Ackva

#### ## Per-Year tCO<sub>2</sub>e Figures for Lifestyle Actions

\*\*Electric Car:\*\* "reduce your emissions from driving from 2.4 tonnes to 1.4 tonnes by switching to an electric car"

\*\*Car-Free Living:\*\* "you could reduce your emissions by around 2.4 tonnes" by going completely without a vehicle  
... [truncated]

#### # Climate Mitigation Potential: Key Consumption Options

Based on the research paper's findings, here are the annual mitigation potentials in tCO<sub>2</sub>e per capita per year:

\*\*Renewable Electricity (Purchasing):\*\* "average values of 1.5 (ranging between 2.5 and 0.3)" with median of 1.6 tCO<sub>2</sub>eq/cap

\*\*Heat Pump:\*\* Average mitigation potential of 0.8 tCO<sub>2</sub>eq/cap

\*\*Renovation/Refurbishment:\*\* "average mit ... [truncated]

I've finished the **carbon** items, the third and fourth groups. The result is stark: many of the current figures can't be traced to a source that says what the page claims, and several differ from what their cited source actually says.

**Sources verified directly:** - EPA's page on a typical passenger vehicle - Wynes & Nicholas (2017), *Environmental Research Letters* - Ivanova et al. (2020), *Environmental Research Letters* - Founders Pledge (2020) - Poore & Nemecek (2018), via Our World in Data's dataset - Wells et al. (2012), *Journal of Industrial Ecology*, abstract via Crossref - Levi's jeans life-cycle assessment (2015) - Sony's PS5 figures (S33)

**Daily carbon items:**

| Item                                                                      | Current | Verified                                                                                               | Status                           |
|---------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------|----------------------------------|
| A cup of coffee                                                           | 0.21 kg | <b>0.20 kg</b> (28.5 kg/kg × 7 g, Poore & Nemecek; farm to retail)                                     | correct it                       |
| An hour on a PS5                                                          | 0.08 kg | <b>0.216 kWh × the region's grid</b> (S33)                                                             | keep, now tied to the grid       |
| A mile in a gas car                                                       | 0.40 kg | <b>0.40 kg</b> , tailpipe CO <sub>2</sub> only (EPA)                                                   | keep, with the boundary labelled |
| Printing a paperback book                                                 | 2.7 kg  | <b>2.71 kg</b> , cradle to gate (Wells et al.)                                                         | keep                             |
| A beef burger (113 g)                                                     | 3.0 kg  | <b>11.2 kg</b> from beef-herd beef (99.5 kg/kg); 3.8 kg from dairy-herd beef                           | <b>correct it</b> ; see below    |
| Microwave, 10-min hot shower, dishwasher, dryer, hour of air conditioning | various | the cited pages don't give the kWh used, or block access; the shower depends on an assumed heater type | <b>remove</b>                    |

**Yearly "add" items:** - **Jeans:** 33 kg is correct (Levi's: 33.4 kg over the life cycle), so it stays. - **Transatlantic round trip:** 1.6 t is correct (Wynes & Nicholas and Founders Pledge), so it stays. - **A year of driving:** 4.8 t becomes **4.6 t** (EPA: 11,500 miles, tailpipe only). - **A year of daily coffee:** 77 kg becomes **about 73 kg**. - **A burger every week:** 156 kg becomes **about 580 kg**. - **Removed:** the T-shirt, sofa, bicycle and car-manufacturing figures cite *How Bad Are Bananas?*, and the book's website gives no figures. The short-haul and US cross-country flights cite an interactive calculator whose outputs I can't verify.

**Yearly "cut" items:** - **Car-free:** 2.4 t is correct (Wynes & Nicholas), so it stays. - **Hang-drying:** 140 kg becomes **210 kg** (Wynes & Nicholas). - **Recycling:** 140 kg becomes **210 kg** (Wynes & Nicholas). - **Hybrid car:** 480 kg becomes **700 kg**, with a range of −0.2 to 3.1 t (Ivanova). - **Green electricity:** 1.38 t becomes **1.5 t**, range 0.3–2.5 t (Ivanova). - **Electric car:** 1.38 t becomes **2.0 t**, range −1.9 to 5.4 t (Ivanova). Wynes & Nicholas give 1.15 t, which would be disclosed. - **Vegan diet:** 1.45 t becomes **0.9 t** (Ivanova). - **Deep retrofit:** 1.58 t becomes **0.9 t** (Ivanova). - **Heat-pump heating:** 1.99 t becomes **0.8 t** (Ivanova). - **Removed:** LED bulbs, because Wynes & Nicholas explicitly don't quantify them.

**The burger is the one real choice.** Poore & Nemecek report beef from dedicated beef herds (99.5 kg CO<sub>2</sub>e/kg) and from dairy herds (33.3 kg/kg) separately. I'd use **beef-herd** beef as the central value and show dairy-herd beef as the low end, because most burger beef comes from beef herds. That's my reasoning, though, not a sourced share.

**Taken together:** the daily carbon chart shrinks to five items. The burger roughly quadruples, and most "cut" figures fall. These changes move comparisons in both directions, which is the point of the feature.

**Question:** Do you accept these carbon corrections and removals, including the beef-herd basis for the burger?

## 50. Learner (RG Sureshkumar)

accept the corrections and removals

## Agent

[Tool: Edit — research.md]

The file research.md has been updated successfully.

[Tool: Bash — Complete feature 5 draft and check for open items]

(Bash completed with no output)

Feature 5 is now fully drafted in [spec.md](#), under **Features** → **Feature 5**, with no open items. All the corrected tables are in, along with the burger's water range (UK about 67 L/kg to US irrigated about 2,000 L/kg, with the global figure of 550 in between).

**Eight new sources added:** S73–S80 are in research.md, marked not yet assessed. There's also a note listing the sources that failed verification, so the removals have a record.

Please check these proposals of mine, which you haven't decided explicitly: - **Labels:** each item shows its unit, assumption and boundary (for example "per 113 g patty · farm to retail") in its label or tooltip. Your rule requires a *stated* boundary. I've put it in the interface, not only in the methodology text. - **Ranges:** items with a reported range (the burger and the Ivanova cuts) show that range. - **Methodology rewrite:** it explains in plain language why a burger's total water (about 15,000 L/kg) differs from its blue water.

**Question:** Does the Feature 5 draft look right? Once you confirm, the remaining steps are the **Out of scope** section and a final read-through of the whole spec for your approval.

## 51. Learner (RG Sureshkumar)

---

feature 5 is good, lets move to the out of scope section

## Agent

---

[Tool: Edit — spec.md]

The file spec.md has been updated successfully.

I've drafted **Out of scope** in [spec.md](#). It has three groups:

1. **Alternatives you rejected when selecting features:** video generation, ranges carried through every result, the water boundary selector, a separate disclosure panel, AI as a share of the digital day, attempts per output, and a US grid selector.
2. **Things excluded by decisions in this stage:** for example image model choice, per-model input-token rates, cloud gaming, digital-day figures in the budget and team view, manufacturing energy, and the removed comparison items.
3. **Things this project doesn't change:** the EcoLogits text-prompt figures and the yearly baseline footprint settings.

While writing group 3, I found an inconsistency you should decide on. The baseline's **driving setting** ("an average amount") uses **4,800 kg a year** (0.40 kg × 12,000 miles). Feature 5 corrects the "year of driving"

comparison to **4.6 t** (EPA: 11,500 miles). Left alone, the same page would give two different figures for a year of average driving.

**Question:** Should the baseline driving setting be brought into line with EPA (4,600 kg for "an average amount", the other options scaled to match)? Or should it stay out of scope, with the difference noted?

## 52. Learner (RG Sureshkumar)

---

bring it in line with EPA

### Agent

---

[Tool: Bash — Record driving baseline decision in spec]

ok

I've recorded the driving decision under feature 5. Next I'm reading the whole spec through to check it's consistent before you review it.

[Tool: Read — spec.md]

```
1 # Technical Specification
2
3 > EDITING DIRECTIVE: USER AND AGENT EDIT THIS FILE COLLABORATIVELY. THE USER MUST
 REVIEW AND APPROVE ITS CONTENT.
4
5 Purpose of this file: Define what the completed project must do so it can be planned,
 built, and verified.
6
7 ## Instructions for the user
8
9 Translate the approved research into a specification without distorting its evidence,
 limitations, or ... [truncated]
```

[Tool: Bash — Fix stale wording and add feature 5 revision note]

done

I've read the whole spec through and made three small fixes: - **Stale wording.** Feature 5's "current state" list still said the items "have not been checked". It now says they were checked afterwards. - **Formatting.** I removed a stray blank line. - **A third Revisions note.** It records that you widened feature 5 beyond the approved "water like for like" wording: carbon, both totals, the verification rule, and the driving baseline.

**The spec in [spec.md](#) is complete, with no open items.** It contains: - **Goal:** two totals side by side, the AI total and the digital-day total. - **Features 1–5:** each with its need, behaviour, calculation, limitations, interface expectations and numbered acceptance checks. - **Out of scope and Revisions:** three revision notes, for features 1, 2 and 5.

Before you decide, check these four things yourself:

1. **Items I proposed that you accepted without an explicit decision.** They're marked "proposed by the agent" in the text:
2. the output-token model choice (feature 2)
3. the device power table and the activity–device combinations (feature 3)
4. device rows 1–5 and the replacement of the old device comparison figures (feature 4)
5. **Sources S59–S80 in research.md** were added during this stage and are marked "not yet assessed by the user". The spec depends on them, so please review their entries.
6. **The S7 discrepancy is still in the approved research.** research.md §3 says agentic coding uses "about 1,000×" more tokens. S7 actually says 3,500× versus single-round reasoning and 1,200× versus multi-round chat. It's your call whether to correct it.
7. **All my arithmetic in the acceptance checks** (for example 46.7 Wh per session, 366 Wh for the digital day, 162 g per day for manufacturing) was calculated from the decided figures. It's worth spot-checking.

**Question:** After reviewing [spec.md](#), do you approve the specification, or is there anything you want changed first? The approval has to come from you. I won't mark it approved.

## 53. Learner (RG Sureshkumar)

---

i approve the specification

## Agent

---

[Tool: Bash — Record user approval and remove draft markers]

```
7
M research.md
M spec.md
```

I've recorded your approval in the **User approval** section of [spec.md](#), with your words and today's date, and removed the seven "draft, not approved" markers. The record also notes that sources S59–S80 in research.md were still marked "not yet assessed" when you approved, so that's on file for later stages.

[spec.md](#) and [research.md](#) are changed but not committed. I'll commit only when you ask.

The specification transcript is a required deliverable for this stage. When you're ready, say **save transcript** and I'll save the whole conversation to transcripts/spec-YYYY-MM-DD\_HHMMSS.md.

## 54. Learner (RG Sureshkumar)

---

save transcript

## Agent

---

[Tool: Bash — Read transcript conventions and research transcript format]

[Tool: Read — ]

```
<system-reminder>This memory is 14 days old. Memories are point-in-time observations, not live state – claims about code behavior or file:line citations may be outdated. Verify against current code before asserting as fact.</system-reminder>
```

```
1 ---
2 name: transcript-export-as-pdf
3 description: "The learner wants session transcripts exported as PDF, not only as Markdown."
4 metadata:
5 node_typ ... [truncated]
```

Exit code 1

# Transcripts

Session transcripts for this project, kept as evidence of process.

- `session-2026-09-12.md` – the build session: porting Andy Masley's calculator to a standalone site and adding the three features.
- `debrief-\*.md` – Unit 1 debrief conversations, saved separately so debrief evidence stays distinct from build sessions.
- `debrief-\*.pdf` – the same debrief rendered f ... [truncated]

[Tool: Bash — Inspect research transcript format and session logs]

# Project 2 Research Stage – transcript

- **Project:** ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator
- **Repository:** <https://github.com/rg-sureshkumar-goat/AI-Footprint-Calculator>
- **Branch:** `project-2` (HEAD at start of session: `d871894`; at save: `d871894`)
- **Date:** 2026-09-26
- **Learner:** RG Sureshkumar
- ... [truncated]

[Tool: Bash — Inspect session log structure]

```
264 ('attachment', None, ())
170 ('assistant', 'assistant', ('tool_use',))
170 ('user', 'user', ('tool_result',))
145 ('assistant', 'assistant', ('thinking',))
108 ('queue-operation', None, ())
75 ('custom-title', None, ())
74 ('last-prompt', None, ())
74 ('agent-name', None, ())
74 ('atis-latch', None, ())
69 ('assistant', 'assistant', ('text',))
54 ('user', 'user', ('str',))
54 ('file-history-sn ... [truncated]
```

*[Tool: Bash — List user text messages in log]*

```
None 'start specification'
None 'I want two totals side by side'
None 'both are right, let move to feature 1'
None 'go with option a'
None 'go with option 1'
None 'go with option a'
None 'go with option b'
None 'go with option a'
None 'looks right, lets move onto feature 2'
None 'go with option c'
None 'go with option 3'
None 'go with option 1'
None 'go with option a'
None 'go with option a'
None ... [truncated]
```

*[Tool: Bash — Check Markdown library and Chrome availability]*

```
Traceback (most recent call last):
 File "<string>", line 1, in <module>
ModuleNotFoundError: No module named 'markdown'
You should consider upgrading via the '/Library/Developer/CommandLineTools/usr/bin/python3
-m pip install --upgrade pip' command.
md ok 3.9
/Applications/Google Chrome.app/Contents/MacOS/Google Chrome
```

I'm writing the script that turns this session's log into the transcript, following the research transcript's format and the sanitising rules in the README.

*[Tool: Write — ]*

```
File created successfully at: <session-temp> (file state is current in your context – no
need to Read it back)
```